

An Assessment of Sampling Error and Variance Estimation Methods: Evidence from Time Use Survey (2024)

By

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A Dissertation submitted for the partial fulfillment of the requirement for the award of
Masters in Survey Research and Data Analytics (2024-26)

Under the guidance of

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(स्थापना/ Established in 1956)
बेहतर भविष्य के लिए क्षमता निर्माण
Capacity Building for a Better Future

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अन्तर्राष्ट्रीय जनसंख्या विज्ञान संस्थान

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स्वास्थ्य एवं परिवार कल्याण मंत्रालय, भारत सरकार का स्वायत्त संगठन
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Capacity Building for a Better Future

International Institute for Population Sciences

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DECLARATION

I hereby declare that the work presented in this dissertation entitled "*An Assessment of Sampling Error and Variance Estimation Methods: Evidence from Time Use Survey (2024)*" has been carried out entirely by me under the supervision of Prof. Chander Shekhar, Head, Department of Fertility and Social Demography, International Institute for Population Sciences, Deonar, Mumbai, for the degree of Masters of Science in Survey Research and Data Analytics for the Academic Year (2024-26).

I further declare that this work is original and has not been submitted in part or whole to any University/Institute for the award of any Degree/Diploma.

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CERTIFICATE

I certify that the above-mentioned declaration is true to the best of my knowledge and belief.

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CHAPTER 1

INTRODUCTION

1.1. Background

Surveys are one of the most widely used methods of collecting information about a population in a structured and systematic manner. They are especially valuable when it is impractical or impossible to gather data through complete enumeration or direct observation. Surveys allow researchers and policymakers to study large populations efficiently by collecting information from a representative sample, which can then be generalized to the entire population. They are flexible in terms of content and methodology, enabling the collection of quantitative and qualitative data on demographic, social, economic, and behavioural characteristics. Surveys provide several advantages: they can cover geographically dispersed populations, enable the study of multiple variables simultaneously, are cost-effective compared to full censuses, and support evidence-based policymaking by producing reliable estimates for planning and evaluation purposes.

Despite their advantages, surveys are subject to errors that can affect the accuracy and reliability of the results. Sample survey estimates are affected by two main types of errors: non-sampling errors and sampling errors (Sukhatme, 1954), which together contribute to the total survey error. Non-sampling errors occur from mistakes made during data collection and processing, such as coverage errors, non-response, misinterpretation of questions, and recording or coding errors, and are difficult to quantify statistically. In contrast, sampling errors arise because only a subset of the target population is surveyed, and different samples drawn from the same population would yield different estimates. Sampling error is measurable statistically through standard errors, confidence intervals, and design effects, and estimating it accurately enables researchers to quantify the precision and uncertainty of survey estimates. In several studies, resampling methods such as jackknife and bootstrap techniques have been studied extensively to provide robust variance estimates across a range of estimators and designs (Rao 2009; Beaumont & Emond 2022), and analytical improvements in linearization variance estimation have also been proposed to make variance estimators more accurate for nonlinear statistics (Demnati & Rao, 2004). These methodological contributions underscore the importance of reducing sampling error not only to enhance the reliability of estimates but

also to support meaningful comparisons across populations and improve the quality of evidence used in research and policymaking.

India conducts several major surveys, including the National Sample Survey (NSS), Demographic and Health Surveys (DHS), and Economic Census, which provide critical information on population characteristics, labour force participation, household consumption, health, and education. These surveys are carefully designed to be representative at national and state levels, using stratified multi-stage sampling and weighting procedures to reduce bias. Like all sample surveys, they are subject to sampling error, which arises because only a subset of the population is observed. Survey agencies address this by employing rigorous variance estimation techniques to provide reliable standard errors, confidence intervals, and design effects. These practices ensure the quality and reliability of survey estimates, allowing researchers and policymakers to draw meaningful conclusions.

In this study, the Time Use Survey (TUS) 2024 (MoSPI, 2025) is used as the primary data source. The TUS specifically captures detailed information on how individuals allocate their time to various activities, including paid employment, unpaid domestic work, education, and leisure. This makes it an ideal dataset for studying patterns of time use and analysing statistical properties such as sampling error. Using TUS as a representative dataset allows the methods and findings to be generalized to other large-scale surveys, making the study relevant for a broad range of survey-based research in India and elsewhere.

The present study aims to evaluate and compare variance estimation techniques under both design-based and model-based approaches. The design-based approach relies on the survey's sampling structure and includes methods such as Taylor Series Linearization, Jackknife Repeated Replication, Balanced Repeated Replication, and Bootstrap replication, which take into account stratification, clustering, and unequal probabilities of selection. The model-based approach, on the other hand, uses statistical modelling frameworks, such as Weighted Least Squares (WLS) regression and bootstrap-based regression, to estimate variances by assuming a model for the underlying data. By applying these methods to the TUS data, this study assesses measures of sampling error, including relative standard error (RSE) and design effect (DEFF), providing a comprehensive understanding of the precision, reliability, and uncertainty of survey estimates. The insights gained can help improve survey analysis, strengthen policy decisions, and serve as a reference for future survey-based research.

1.2. Literature Review

Survey sampling and variance estimation have been extensively studied, beginning with foundational contributions by Cochran (1977), who outlined procedures for variance estimation under complex survey designs and highlighted the limitations of simple random sampling. Kish (1965) introduced the concept of design effects, demonstrating how complex sampling structures influence variance and emphasizing the importance of appropriate adjustments in statistical inference. These works established the theoretical basis for subsequent developments in variance estimation methods.

Taylor series linearization is one of the classical approaches for estimating the variance of complex statistics. Woodruff (1971) formalized its application in surveys by approximating nonlinear statistics as linear functions, enabling the use of standard variance formulas. Rao and Wu (1988) extended this framework to more complex estimators, such as ratios and regression coefficients, while noting that accuracy depends on the smoothness of the estimator. Sarndal et al. (1992) highlighted its practical importance in survey analysis, while also acknowledging potential limitations under highly complex designs.

As an alternative, the Jackknife method originally introduced by Quenouille (1949) for bias reduction and later adapted for variance estimation by Tukey (1958) and formalized for complex survey applications by Durbin (1959) estimates sampling variance by systematically deleting one sampling unit at a time and comparing replicate estimates to the full-sample estimate. Rao and Shao (1999) demonstrated that the jackknife provides reliable and approximately unbiased variance estimates across different sampling schemes, including stratified and unequal probability designs, though at a higher computational cost as sample size and design complexity increase. The Bootstrap method, introduced by Efron (1979) and later adapted for complex surveys by Rao and Wu (1988), offers greater flexibility across a wider range of design configurations. Paben (2001) evaluated all these methods together, finding that while they produce similar results for large samples, replication-based methods more effectively capture variability in smaller domains where linearization tends to underestimate variance.

Verma and Betti (2011) developed a systematic procedure for estimating sampling errors and design effects for complex poverty and inequality statistics derived from household surveys, deriving explicit linearized variables for a wide range of measures including the Head Count Ratio, FGT indices, Gini coefficient, and Atkinson indices, and demonstrating that the

linearization and jackknife approaches yield broadly similar results, with differences emerging mainly for quantile-based statistics.

In addition to design-based methods, recent research has extended variance estimation to model-based contexts, where both design and model uncertainty must be considered. Romano and Wolf (2017) show that weighted least squares with robust standard errors improves efficiency under heteroskedasticity, while Beaumont and Charest (2012) demonstrate that generalized bootstrap methods can provide accurate variance estimates even when model parameters are estimated from complex survey data.

1.3. Need for the study

Sampling error, which arises inevitably from the process of observing a subset of the population rather than the population in its entirety, represents a central concern in the analysis of complex survey data. The analysis of complex survey data constitutes a fundamental component of empirical research in the social sciences, economics, and public policy. Large-scale national surveys typically employ complex sampling designs that incorporate stratification, clustering, and unequal probabilities of selection in order to ensure adequate representation of diverse population groups. While such designs enhance the efficiency and representativeness of survey estimates, they simultaneously introduce methodological challenges in statistical inference, particularly in the estimation of variances and standard errors. Accurate variance estimation is essential for assessing the precision and reliability of survey estimates. Standard errors and confidence intervals derived from appropriate variance estimation techniques enable researchers to evaluate the statistical uncertainty associated with survey estimates and support sound interpretation of empirical findings. However simple random sampling, where variance estimation follows straightforward analytical formulas, complex sampling designs require more sophisticated methods to accurately quantify sampling error and produce reliable standard errors for estimated parameters.

To address these challenges, several methodological approaches have been developed for variance estimation in complex surveys. Among them, the Taylor Series Linearization (TSL) method is widely used due to its computational efficiency and applicability to a broad class of estimators. In addition, replication-based methods, such as Jackknife, Balanced Repeated Replication (BRR), and Bootstrap techniques, have gained prominence because of their flexibility in handling complex estimators and non-linear statistics. These approaches

collectively constitute the design-based framework, where statistical inference is grounded in the sampling design.

Existing literature on variance estimation in complex survey data has largely concentrated on comparisons among design-based approach, particularly Taylor linearization and replication techniques. However, there is limited empirical evidence on how design-based and model-based approaches differ when applied to real survey data. This gap highlights the need for a study that examines the performance and applicability of both approaches in estimating sampling error.

Despite the extensive use of design-based approaches for variance estimation in complex survey data, relatively limited attention has been given to examining how model-based approaches perform in this context using real survey datasets. In particular, methods such as Weighted Least Squares (WLS) and Bootstrap-based WLS are widely applied in empirical analysis, yet their implications for estimating sampling error in complex survey settings are not fully understood. Moreover, existing studies often emphasize methodological comparisons without adequately exploring the reliability of estimates in terms of measures such as relative standard error (RSE) and design effect (DEFF). Therefore, there is a need to investigate the behaviour of sampling error using both design-based and model-based approaches, and to assess their performance and applicability in the context of complex survey data. This study aims to address this gap using evidence from the Time Use Survey.

The Time Use Survey conducted in 2024 provides a valuable opportunity to explore these methodological issues, as it captures detailed information on time allocation patterns and unpaid activities that are crucial for understanding socio-economic dynamics and gender disparities. In this context, a systematic assessment of variance estimators and confidence intervals derived from both design-based and model-based approaches becomes particularly relevant. Such an analysis facilitates a deeper understanding of sampling error through measures such as standard errors, relative standard error (RSE), and design effect (DEFF), and helps examine the consistency and reliability of estimates produced by these approaches. This, in turn, contributes to the broader methodological discussion on the appropriate treatment of complex survey data. Ultimately, the findings of this study are expected to enhance the reliability and interpretability of survey-based research and support evidence-based policy formulation.

1.4. Research Questions

1. What is the mean time spent on unpaid and Childcare activities among all individuals, leisure and self-care activities among women in India?
2. What are the estimated relative standard errors (RSE) and design effects (DEFF) under Design-based methods, and the relative standard errors (RSE) under Model-based methods for selected indicators?
3. How consistent and reliable are the Sampling Error estimates obtained from Design based and Model based methods?
4. How do different variance estimation methods perform in estimating sampling error in Time Use Survey?

1.5. Research Objectives

1. To estimate mean time spent on Unpaid and childcare activities among all individuals, leisure and self-care activities among women in India.
2. To estimate and evaluate sampling error for Time Use Survey estimates using design-based variance estimation methods and to assess their performance and applicability.
3. To estimate and evaluate sampling error for Time Use Survey estimates using model-based variance estimation methods and to examine their performance and applicability.

1.6. Organization of the Dissertation

- Chapter 1: Introduction
- Chapter 2: Data and Methodology
- Chapter 3: Descriptive Analysis of Time Use Distribution
- Chapter 4: Assessment of Sampling Errors Using Design-Based Variance Estimation Methods
- Chapter 5: Assessment of Sampling Errors Using Model-Based Variance Estimation Methods
- Chapter 6: Conclusion

CHAPTER 2

DATA AND METHODOLOGY

2.1. Data Source:

We illustrate the survey sampling errors and the comparison between measures using the secondary data obtained from the Time Use Survey (TUS), 2024, conducted by the National Statistics Office under the Ministry of Statistics and Programme Implementation (MoSPI), Government of India. The Time Use Survey is a nationally representative survey designed to capture detailed information on how individuals allocate their time across various activities over a 24-hour reference period. The Time Use Survey, 2024 is the second all-India survey in this series, following the first round conducted in 2019. The survey was carried out over a full calendar year (January to December 2024) to account for seasonal variations in time-use patterns.

The Time Use Survey (TUS) follows a stratified two-stage sampling design. In the first stage, villages in rural areas and Urban Frame Survey (UFS) blocks in urban areas were selected as First Stage Units (FSUs). In the second stage, households were selected as Ultimate Stage Units (USUs) using Simple Random Sampling Without Replacement (SRSWOR). A total of 9,969 FSUs were surveyed (5,949 rural and 4,020 urban), covering 1,39,487 households. The survey collected information from individuals aged 6 years and above, with an effective sample of 2,83,015 individuals in rural areas and 1,67,442 individuals in urban areas after excluding non-reporting cases.

Data were collected using a 24-hour time diary method, where each day was divided into 48 time slots of 30 minutes each, starting from 4:00 AM of the previous day to 4:00 AM of the survey day. Respondents reported their activities in each time slot, including simultaneous activities classified as major and minor activities.

The survey captures a wide range of activities, including paid work, unpaid domestic services, caregiving activities, training, volunteering, learning, social participation, and leisure. Activities are classified based on the International Classification of Activities for Time Use Statistics (ICATUS 2016), ensuring comparability at the international level.

The availability of unit-level data and sampling weights makes the dataset suitable for analysing time-use patterns and estimating sampling variability in a complex survey design.

2.2. Indicators construction:

The primary indicators used in this study are constructed from the time-use schedule, which provides detailed information on activities performed by individuals over a 24-hour reference period. Based on this dataset, individual-level measures of time allocation are generated by aggregating the duration (in minutes) spent across different activity categories.

In particular, this study focuses on three key derived indicators: **Total unpaid time, Time poverty rate and Childcare Rate**. Total unpaid time is computed by summing the time spent by individuals on various unpaid activities, such as domestic services and caregiving. The time poverty rate is then derived as a binary indicator based on a predefined threshold of total leisure time, identifying working women who experience time constraints beyond a specified limit. Additionally, the Childcare Rate captures the proportion of elderly individuals engaged in providing care to children, reflecting their contribution to unpaid caregiving within households. For the construction of these variables, the analysis was restricted to major activities to ensure that each time interval is uniquely assigned to a single dominant activity, thereby avoiding overlap and double counting and only data collected on normal days were considered to ensure consistency in measuring routine time use.

These constructed indicators enable the analysis of the distribution of total time allocated to unpaid activities over a 24-hour period, the prevalence of time poverty among working women, and the extent of elderly participation in childcare, forming the basis for subsequent statistical estimation and inference.

2.2.1. Total Unpaid Time

The unpaid activities include the production of goods for own final use, unpaid domestic services for household members, unpaid caregiving services for household members, unpaid volunteer, trainee and other unpaid work.

2.2.1.1. Identification of Relevant Activity Episodes

The construction of total unpaid work time begins with the identification of relevant activity episodes from the time-use schedule. Activities classified as unpaid work are selected based on the activity code criterion:

$$A_{ij} = \begin{cases} 1, & \text{if } 211 \leq \text{Activity Code}_{ij} \leq 590 \\ 0, & \text{otherwise} \end{cases}$$

where A_{ij} indicates whether the j^{th} activity episode of individual i falls within the unpaid work category. (Appendix 1)

2.2.1.2. Conversion of Time into Minutes

The start and end times of each activity episode, originally recorded in 24-hour format, are converted into minutes from midnight:

$$\begin{aligned} t_{ij}^{\text{start}} &= (h_{ij}^{\text{start}} \times 60) + m_{ij}^{\text{start}} \\ t_{ij}^{\text{end}} &= (h_{ij}^{\text{end}} \times 60) + m_{ij}^{\text{end}} \end{aligned}$$

where h and m represent the hour and minute components, respectively.

2.2.1.3. Computation of Activity Duration

The duration of each activity episode is calculated as:

$$d_{ij} = \begin{cases} t_{ij}^{\text{end}} - t_{ij}^{\text{start}}, & \text{if } t_{ij}^{\text{end}} \geq t_{ij}^{\text{start}} \\ (t_{ij}^{\text{end}} - t_{ij}^{\text{start}}) + 1440, & \text{if } t_{ij}^{\text{end}} < t_{ij}^{\text{start}} \end{cases}$$

The adjustment for cases where $t_{ij}^{\text{end}} < t_{ij}^{\text{start}}$ accounts for activity episodes that extend past midnight, ensuring accurate measurement within the 24-hour reference period.

2.2.1.4. Aggregation of Unpaid Time

The unique identifier created using FSU serial number, sample household number and person serial number to convert the activity level to individual level. The total unpaid time for each individual is computed by summing the duration of all qualifying activity episodes:

$$\text{Total Unpaid Time}_i = \sum_{j=1}^{n_i} d_{ij}$$

where the summation is taken over all activity episodes satisfying $A_{ij} = 1$ and $M_{ij} = 1$.

2.2.2 Time Poverty Rate

Time poverty in this study is estimated specifically for working women, in order to capture time constraints among individuals engaged in economic activities. Accordingly, the sample is restricted to women and whose principal activity status indicates participation in work-related categories.

2.2.2.1. Conversion of Time into Minutes and computation of Activity Duration

In the section 2.2.1.2. and 2.2.1.3., the same formula we used here. The recorded start and end times are converted into minutes from midnight and calculated the activity duration.

2.2.2.2. Identification of Leisure Activities

Leisure activities are identified using activity codes: (Appendix 1)

$$L_{ij} = \begin{cases} 1, & \text{if } 700 \leq \text{Activity Code}_{ij} \leq 899 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Leisure Time}_{ij} = L_{ij} \cdot d_{ij}$$

2.2.2.3. Aggregation to Individual Level

$$\text{Total Leisure Time}_i = \sum_{j=1}^{n_i} \text{Leisure Time}_{ij} \text{ and } \text{Total Time}_i = \sum_{j=1}^{n_i} d_{ij}$$

2.2.2.4. Definition of Time Poverty

Time poverty is defined as a binary indicator:

$$\text{Time Poor}_i = \begin{cases} 1, & \text{if } \text{Total Leisure Time}_i \leq 60 \\ 0, & \text{otherwise} \end{cases}$$

Thus, individuals with 60 minutes or less of leisure time per day are classified as time poor. The 60-minute threshold captures severe time constraints by identifying individuals with minimal discretionary time for rest and well-being, which is particularly relevant for working women facing multiple responsibilities.

2.2.2.5. Time Poverty Rate

$$\text{Time Poverty Rate} = \frac{1}{N} \sum_{i=1}^N \text{Time Poor}_i$$

2.2.3. Childcare Rate

The childcare rate measures the participation of elderly individuals (aged 60 years and above) in providing care to children.

2.2.3.1 Conversion of Time into Minutes

In the section 2.2.1.2. and 2.2.1.3., the same formula we used here. The recorded start and end times are converted into minutes from midnight and calculated the activity duration.

2.2.3.2 Identification of Childcare Activities

Childcare activities are identified using 3-digit activity codes (411–419). A binary indicator is defined as: (Appendix 1)

$$\text{ChildcareAct}_{ij} = \begin{cases} 1, & \text{if } 411 \leq \text{ActivityCode}_{ij} \leq 419 \\ 0, & \text{otherwise} \end{cases}$$

The corresponding time spent is:

$$\text{ChildcareTime}_{ij} = \text{Duration}_{ij} \times \text{ChildcareAct}_{ij}$$

2.2.3.3 Aggregation to Individual Level

Total childcare time for individual i is computed as:

$$\text{TotalChildcareTime}_i = \sum_{j=1}^{n_i} \text{ChildcareTime}_{ij}$$

A participation indicator is defined as:

$$\text{ParticipateChildcare}_i = \begin{cases} 1, & \text{if } \text{TotalChildcareTime}_i > 0 \\ 0, & \text{otherwise} \end{cases}$$

2.2.3.4 Childcare Rate

The childcare participation rate among elderly individuals is given by:

$$\text{ChildcareRate} = \frac{1}{N} \sum_{i=1}^N \text{ParticipateChildcare}_i$$

This indicator reflects the prevalence of childcare responsibilities undertaken by elderly individuals in the surveyed population.

2.3. Background Characteristics of Study Population:

<i>Variable</i>	<i>Categories</i>
<i>Gender</i>	Male, Female
<i>Employment Status</i>	Employed, Unemployed, Not in Labour Force (NILF)
<i>Sector</i>	Rural, Urban
<i>Education Level</i>	Non-literate, Primary, Secondary, Graduation
<i>Wealth Index</i>	Poorest, Poor, Middle, Rich, Richest

<i>Social Group</i>	SC, ST, OBC, Others
<i>Religion</i>	Hindu, Islam, Christianity, Others
<i>Household size</i>	Small (1-3), Medium (4-5), Large (6+)
<i>No.of.Children</i>	No children, Single child, 2+ Children
<i>Marital Status</i>	Never married, currently married, Widowed, Divorced/Separated

2.4. Sample Size

The effective sample for the present study is derived after applying the specified selection criteria to the Time Use Survey, 2024 data.

<i>Measure</i>	<i>Sample Size</i>
<i>Total Unpaid Time</i>	3,97,018
<i>Time Poverty Rate</i>	43,309
<i>Childcare Rate</i>	52,544

2.5. Statistical Analysis:

2.5.1. Design Based approaches

2.5.1.1. Taylor Series Linearization Method:

The Taylor Series (TS) Linearization method is a widely used design-based approach for variance estimation in complex surveys. It extends standard variance estimation techniques applicable to linear estimators to nonlinear estimators by approximating them using a first-order Taylor series expansion.

The key idea is to express a nonlinear estimator as an approximate linear function of the observed data. Once linearized, standard variance estimation formulas for linear statistics can be applied.

Consider a stratified sample with strata $h = 1, 2, \dots, H$ and $i = 1, 2, \dots, n_h$ units within a stratum h . Let x_{hi} represent the value of the study variable and w_{hi} be the sampling weight, in i^{th} unit of the h^{th} strata.

Then, estimator of the population total, $\hat{\theta}$ is:

$$\hat{\theta} = \sum_{h=1}^H \sum_{i=1}^{n_h} w_{hi} x_{hi}$$

Under the Taylor Series method, the variance of $\hat{\theta}$ is estimated as:

$$\hat{v}(\hat{\theta}) = \sum_{h=1}^H \frac{n_h}{n_h - 1} \sum_{i=1}^{n_h} (\hat{\theta}_{hi} - \bar{\theta}_h)^2$$

where:

$$\hat{\theta}_{hi} = w_{hi} x_{hi}$$

$$\bar{\theta}_h = \frac{1}{n_h} \sum_{i=1}^{n_h} \hat{\theta}_{hi}$$

For nonlinear estimators (such as ratios, means, or proportions), a first-order Taylor series expansion is used to derive a linearized variable. The variance is then estimated by applying the above stratified variance formula to the linearized values.

2.5.1.2. Jackknife Repeated Replication:

Jackknife Repeated Replication (JRR) is a replication-based method widely used for estimating sampling variance in complex survey data. The method works by generating multiple replicate samples and measuring the variability among the corresponding estimates.

In complex survey designs, JRR is implemented by systematically deleting one Primary Sampling Unit (PSU) at a time within each stratum and adjusting the weights of the remaining units. Each replicate reflects the structure of the full sample while incorporating the effect of excluding one PSU.

Let $\hat{\theta}$ denote the estimate of the parameter of interest based on the full sample, and let $\hat{\theta}_{(kj)}$ denote the estimate obtained after deleting the j -th PSU from the k -th stratum.

Weight Adjustment:

$$g_{kj} = \frac{w_k}{w_k - w_{kj}}$$

where:

$w_k = \sum_j w_{kj}$, and w_{kj} is the weight of PSU j in stratum k .

Adjusted weights:

- For units not in stratum k :

$$w_{kji}^* = w_{kji}$$

- For units in stratum k but not in PSU j :

$$w_{kji}^* = g_{kj} \cdot w_{kji}$$

- For units in PSU j :

$$w_{kji}^* = 0$$

Mean of Replicate Estimates,

$$\hat{\theta}_{(k)} = \frac{1}{a_k} \sum_{j=1}^{a_k} \hat{\theta}_{(kj)}$$

The JRR Variance Estimate is given by:

$$\widehat{V}_{JRR}(\hat{\theta}) = \sum_k (1 - f_k) \frac{a_k - 1}{a_k} \sum_{j=1}^{a_k} (\hat{\theta}_{(kj)} - \hat{\theta}_{(k)})^2$$

$k \rightarrow$ stratum, $j \rightarrow$ PSU within stratum, $a_k \rightarrow$ number of PSUs in that stratum, $f_k \rightarrow$ sampling fraction

2.5.1.3. The Bootstrap Method:

The Bootstrap method is a widely used resampling technique for estimating the variance and standard error of complex estimators. It is closely related to replication-based methods such as Jackknife Repeated Replication (JRR). The basic idea is to generate multiple replicate samples from the observed data and recompute the statistic of interest for each replicate.

In the context of complex survey data, bootstrap replicates are created by repeatedly resampling Primary Sampling Units (PSUs) within each stratum, maintaining the structure of the original sampling design. During resampling, some PSUs may be selected more than once, while others may not be selected. This process is repeated a large number of times, say B times (typically $B \geq 200$), to ensure stable variance estimates.

Let $\hat{\theta}$ be the estimate based on the full sample, and let $\hat{\theta}^{(b)}$ denote the estimate from the b -th bootstrap replicate, where $b = 1, 2, \dots, B$. The bootstrap estimate of variance is given by:

$$\hat{V}_{boot}(\hat{\theta}) = \frac{1}{B} \sum_{b=1}^B (\hat{\theta}^{(b)} - \bar{\theta})^2$$

where:

$$\bar{\theta} = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^{(b)}$$

In practice, a bias correction factor $\frac{n-1}{n}$ may be applied, especially when the sample size is small. Alternatively, in stratified designs, a common approach is to resample $(n_h - 1)$ PSUs from each stratum h , and adjust the sampling weights of selected units by a factor of $\frac{n_h}{n_h - 1}$ to maintain design consistency.

The Bootstrap method is particularly useful for non-linear estimators, where analytical variance estimation (such as Taylor linearization) becomes difficult. However, it is computationally intensive and requires a sufficiently large number of replicates to obtain stable estimates.

2.5.2 Model based approaches

2.5.2.1. Ordinary Least Squares (OLS) Regression

Ordinary Least Squares (OLS) regression is a fundamental statistical method used to estimate the relationship between one or more independent variables and a dependent variable. OLS aims to find the best-fitting line (or hyperplane, in the case of multiple predictors) by minimizing the sum of squared differences between the observed values and the predicted values. This method provides unbiased and efficient estimates under the classical assumptions, including linearity, independence, and constant variance of errors.

The OLS estimator of the regression coefficients is given by:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Where $\hat{\beta}$ is the vector of estimated coefficients, X is the matrix of independent variables (including a column of ones for the intercept), and y is the vector of observed values of the dependent variable.

The predicted values of the dependent variable are calculated as:

$$\hat{y} = X\hat{\beta}$$

OLS minimizes the sum of squared residuals, ensuring the model provides the best linear fit to the observed data.

2.5.2.2. Weighted Least Squares (WLS) Regression:

Weighted Least Squares (WLS) regression is an extension of the Ordinary Least Squares (OLS) method used when the assumption of constant variance of the error terms (homoscedasticity) is violated. In the presence of heteroscedasticity, where the variance of the error terms differs across observations, WLS provides more efficient and unbiased estimates by assigning weights to each observation.

In WLS regression, each observation is weighted inversely proportional to the variance of its error term. Observations with lower variance (i.e., more reliable information) are assigned higher weights, while those with higher variance receive lower weights. This ensures that observations with greater precision have a stronger influence on the estimated regression coefficients.

The WLS estimator of the regression coefficients is given by:

$$\hat{\beta} = (X^T W X)^{-1} X^T W y$$

Where $\hat{\beta}$ is the vector of estimated regression coefficients, X is the matrix of independent variable, W is the diagonal matrix of weights and y is the vector of dependent variable. In practice, the weights are typically defined as:

$$w_i = \frac{1}{\sigma_i^2}$$

where σ_i^2 represents the variance of the error term for observation i .

In this study, WLS regression is used as a model-based approach to estimate their variability in the presence of heteroscedasticity. By incorporating weights, the method improves the efficiency of estimates compared to ordinary least squares, particularly when dealing with survey data where variability may differ across observations.

2.5.2.3. Bootstrap Method with Weighted Least Squares (WLS):

The bootstrap method is a resampling-based technique used to estimate the variability of statistical estimators. When combined with Weighted Least Squares (WLS) regression, it

provides a robust approach for estimating the standard errors of regression coefficients, particularly in the presence of heteroscedasticity.

In this approach, repeated bootstrap samples are drawn from the observed dataset with replacement. For each bootstrap sample, the WLS regression model is estimated, and the corresponding regression coefficients are obtained. This process is repeated a large number of times, say B replications.

For each bootstrap sample (b) , the WLS estimator is given by:

$$\hat{\beta}^{(b)} = (X^{(b)T} W^{(b)} X^{(b)})^{-1} X^{(b)T} W^{(b)} y^{(b)}$$

Where:

$X^{(b)} y^{(b)}$ = bootstrap sample data

$W^{(b)}$ = diagonal matrix of weights for bootstrap sample

$\hat{\beta}^{(b)}$ = coefficient estimate from b -th replicate

Bootstrap Variance of WLS Estimator

Let $\hat{\beta}$ be the WLS estimate from the full sample. The bootstrap variance is:

$$\hat{V}_{boot}(\hat{\beta}) = \frac{1}{B} \sum_{b=1}^B (\hat{\beta}^{(b)} - \bar{\hat{\beta}})^2$$

where:

$$\bar{\hat{\beta}} = \frac{1}{B} \sum_{b=1}^B \hat{\beta}^{(b)}$$

The bootstrap method captures the variability of the WLS estimator by repeatedly re-estimating the regression model on resampled datasets. The variation among the bootstrap estimates $\hat{\beta}^{(b)}$ reflects the sampling variability of the estimator, providing an empirical estimate of its standard error. In this study, bootstrap WLS regression is used as a model-based approach to estimate the standard errors of regression coefficients.

2.6. Performance and Applicability of Design Based and Model Based Variance estimation Methods

Design Based Methods	Performance and Applicability
Taylor Linearization	- Works well for large samples. - Can handle stratified and clustered sampling designs. - Applicable for nonlinear estimators if a linearized form can be derived. - Provides efficient variance estimates for standard survey indicators. - Only works if a linearized variable can be derived; sometimes linearization is impossible. - Suitable for averages and totals; not recommended for rate variables when linearization is difficult.
Jackknife Method	- Replication-based; no mathematical approximation needed. - Can be applied to any complexity without deriving a linearized variable. - Works best with more than 2 clusters per stratum and without replacement sampling of clusters. - Performs well for small samples, complex statistics, and rate variables. - Useful for international comparisons and when the statistic is not smooth. - Flexible and robust for complex survey designs. - Suitable for averages, rates, and nonlinear or complex statistics.
Bootstrap Method	- Non-parametric resampling method and Clusters are drawn with replacement, using cluster-level data for variance approximation. - Can be applied to various statistical problems, including nonlinear and complex statistics.

	- Particularly effective for small samples and rate variables. - Computationally heavier than other methods. - Flexible and robust for complex survey designs. - Differs from jackknife in asymptotic behaviour, computational requirements, and practicality. - Suitable for averages, rates, and complex statistics, especially when analytical formulas are unavailable.
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Model Based Methods	Performance and Applicability
Weighted Least Squares (WLS)	- Incorporates survey weights to produce unbiased estimates of means or regression parameters. - Intercept-only WLS provides the weighted mean, offering precise and stable estimates of averages. - Assumes correct model specification, linearity between predictors and outcomes, independence of residuals, homoscedasticity, and normality of errors for inference. - Computationally straightforward and efficient. - Suitable for Averages, totals, and continuous outcomes.
Bootstrap WLS	- Combines WLS with non-parametric resampling, repeatedly resampling the data B times and refitting the model to empirically estimate standard errors and confidence intervals. - Captures additional variability not reflected by standard WLS. - Particularly effective for small samples or when survey weights vary substantially. - Assumes same regression assumptions as WLS additionally the resampling reflects the underlying population.

	- Computationally more intensive than standard WLS due to multiple replicates. - Particularly suitable for averages, totals, continuous outcomes, and rate-type statistics when modelled appropriately.
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2.7. Performance Metrics:

Standard error (SE):

The standard error is the square root of the estimated variance of the estimator and measures the variability of the estimate across samples.

$$SE(\hat{\theta}) = \sqrt{V(\hat{\theta})} \text{ or } SE(\hat{\beta}) = \sqrt{V(\hat{\beta})}$$

Relative Standard Error (RSE):

RSE measures the standard error relative to the estimate and is often expressed as a percentage to assess the precision of the estimate.

$$RSE(\hat{\theta}) = SE(\hat{\theta}) / \hat{\theta} \times 100 \text{ or } RSE(\hat{\beta}) = SE(\hat{\beta}) / \hat{\beta} \times 100$$

Design Effect (DEFF):

A most useful concept for the computation, analysis and interpretation of sampling errors concerns ‘design effect’. Design effect is the ratio of variance $\text{Var}(\hat{\theta})$ under the given sample design, to variance $\text{Var}_{\text{SRS}}(\hat{\theta})$ under a simple random sample of the same size:

$$\text{DEFF}(\hat{\theta}) = \text{Var}(\hat{\theta}) / \text{Var}_{\text{SRS}}(\hat{\theta}), \text{ or } \text{DEFF}(\hat{\beta}) = \text{Var}(\hat{\beta}) / \text{Var}_{\text{SRS}}(\hat{\beta})$$

2.8. Software Support: Statistical analyses were conducted using Stata 17.0, Spyder (Python), and R. Microsoft Excel and Microsoft Word were used for data handling and documentation.

CHAPTER 3

DESCRIPTIVE ANALYSIS OF TIME USE DISTRIBUTION

3.1. Introduction

Time is a limited and valuable resource, and the way people spend their time across different activities reflects important socio-economic patterns in society. Studying how time is allocated helps us better understand everyday life, especially the balance between paid work, unpaid activities, self-care, and leisure. In this context, looking at the distribution of time use is important for identifying inequalities and differences in how various groups of people spend their time.

This chapter provides a descriptive overview of time use patterns using data from the Time Use Survey. The main focus is on the distribution of time spent on unpaid activities and leisure, which are often overlooked but form a significant part of daily life. By examining how these activities are distributed across individuals, the study highlights the differences and disparities that exist within the population. At the same time, it is important to note that these findings are derived from sample data. The distributions presented here not only describe time use patterns but also provide an initial indication of the consistency and stability of the results across different population groups.

The chapter further looks at how time use varies across different socio-economic and demographic groups such as gender, household size, employment status, sector, education, wealth index, and marital status. Analysing these patterns helps in identifying groups that carry a heavier burden of unpaid activities, while also reflecting the quality of the data. For example, unusual patterns or extreme values in the distribution may indicate real differences in behaviour, but they may also arise due to sampling fluctuations.

Overall, this chapter acts as a starting point for the study by providing a clear picture of how time is distributed across activities and groups. It also prepares the ground for the later chapters, where issues related to sampling error and statistical reliability are examined in more detail. The findings from this section therefore play an important role in guiding the further analysis in the study.

3.2. Overall Distribution of Time Use

Time use refers to the way individuals allocate their time across various activities within a given period, typically a day. Understanding the overall distribution of time use provides insights into daily routines, social behaviour, and the balance between work, leisure, and household responsibilities. It helps identify patterns of time allocation among different population groups, including variations by age, gender, occupation, and socio-economic status.

Analysing the overall distribution also highlights key aspects such as unpaid work, care responsibilities, and time poverty, which have important implications for economic productivity and social well-being. By examining the spread and concentration of time across activities, it becomes possible to understand how individuals allocate their daily time and the extent of variation across different groups, thereby providing deeper insights into patterns of time use and everyday life.

Figure 3.2.1 shows the distribution of time across major activities per person, indicating that self-care occupies the largest share of the day, reflecting the essential nature of activities such as sleep and personal maintenance. Leisure constitutes a moderate portion of time use, while paid and unpaid work are relatively comparable, highlighting the substantial role of unpaid activities in daily life. In contrast, learning accounts for the smallest share. Overall, the pattern indicates the dominance of necessary activities alongside a balanced allocation between work-related and leisure activities. Also, reveals a clear gender difference in the distribution of time across activities. Men devote considerably more time to paid work, whereas women spend significantly more time on unpaid domestic activities. When total activity is considered, women combined paid and unpaid activity slightly exceeds that of men, suggesting that the disparity lies not in the total amount of activity performed but, in its distribution, and economic recognition. In addition, time allocated to leisure, learning, and self-care remains broadly similar across genders. Overall, gender disparities in time use are concentrated between paid and unpaid Activity.

Figure 3.2.1 Average Time Allocation Across Activities by Sex

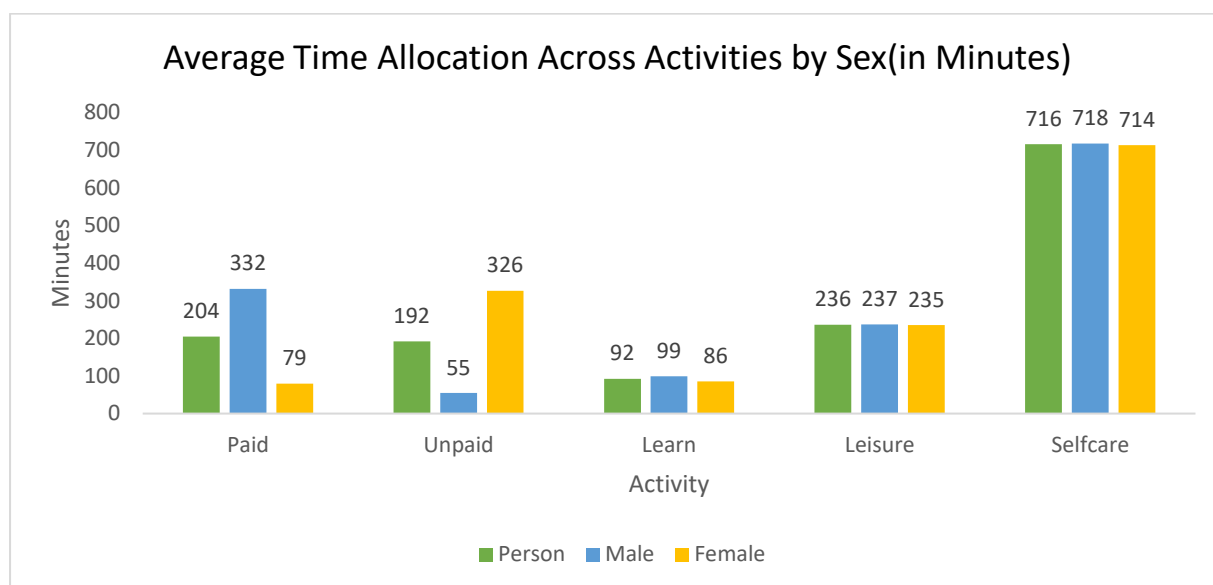


Figure 3.2.2 Time use varies clearly by employment status. Employed individuals spend most of their time in paid activity with limited leisure, while unemployed individuals have more leisure time. Those not in the labour force (NILF) spend almost no time in paid activity but devote a large share of time to unpaid domestic activities, highlighting the hidden burden of unpaid activity.

Figure 3.2.3 Time use patterns are broadly similar across rural and urban areas. Urban individuals spend slightly more time in paid activity, while rural individuals allocate somewhat more time to unpaid activities. Overall, the differences are minimal, indicating that location has a limited impact on daily time allocation.

Figure 3.2.2 Average time spent by Employment Status

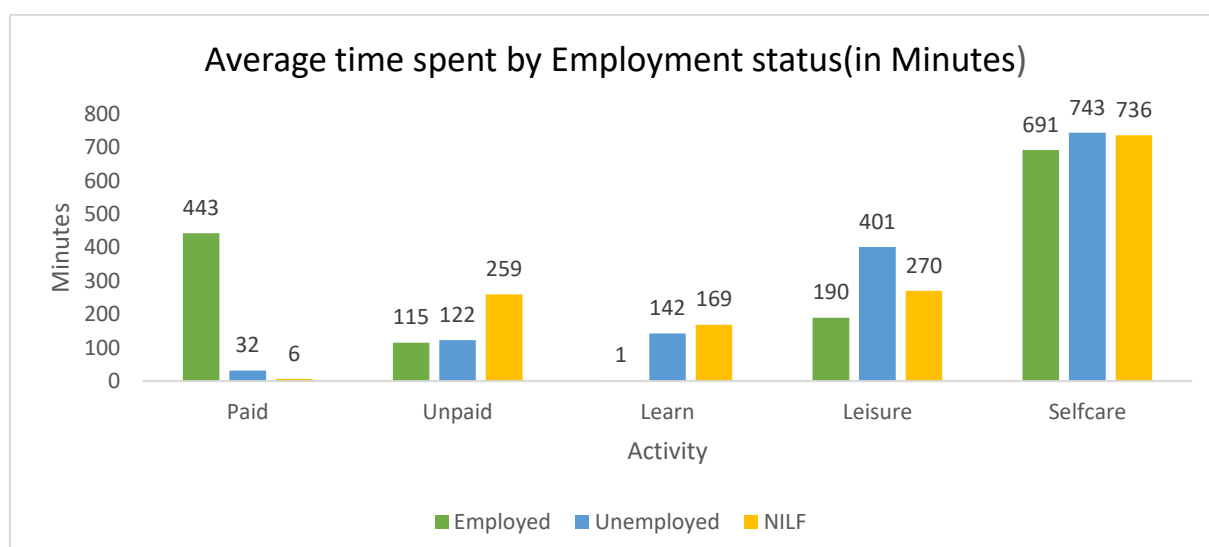


Figure 3.2.3 Average time spent by Sector-wise

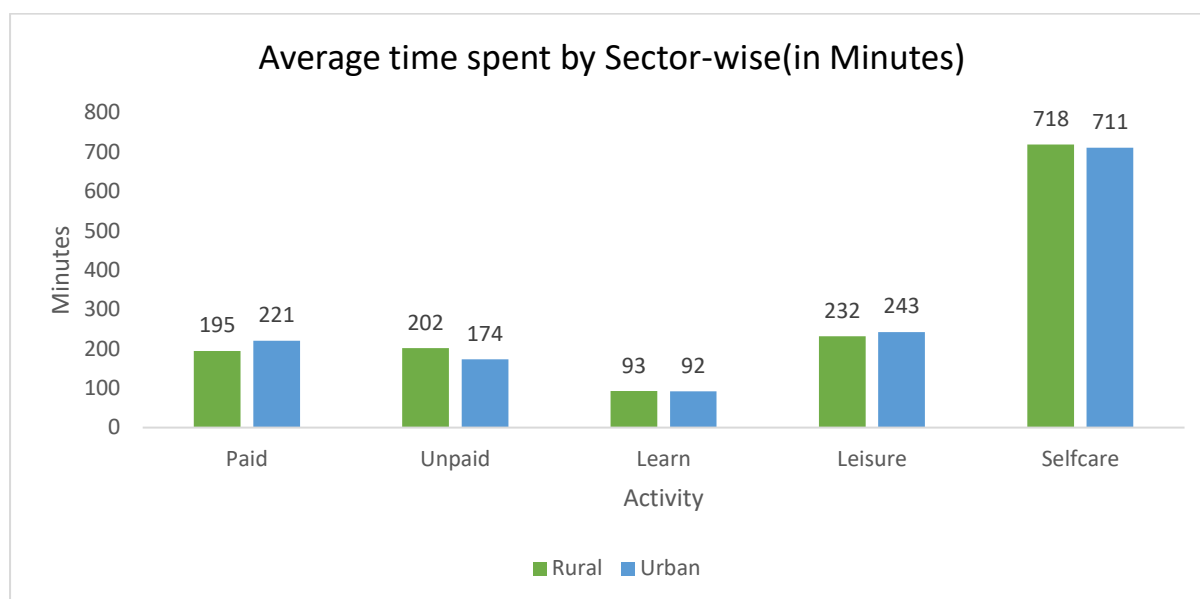


Figure 3.2.4 Time use varies across wealth groups. Higher-wealth individuals spend more time in paid activity, while lower-wealth groups devote more time to unpaid domestic activities, indicating an economic influence on work type. Differences in learning are minor, and leisure and self-care are largely similar, showing that wealth mainly affects work distribution rather than total time use.

Figure 3.2.5 Time use patterns vary across education levels. Higher-educated individuals spend more time in paid activity and less in unpaid domestic activities, while those with lower education dedicate more time to unpaid work and less to learning. Learning time rises with basic education but declines at higher levels, reflecting life stage and work commitments. Self-care shows minor variation, with less educated individuals spending slightly more time. Overall, education strongly influences the distribution and nature of daily activities.

Figure 3.2.4 Average time spent by Wealth index

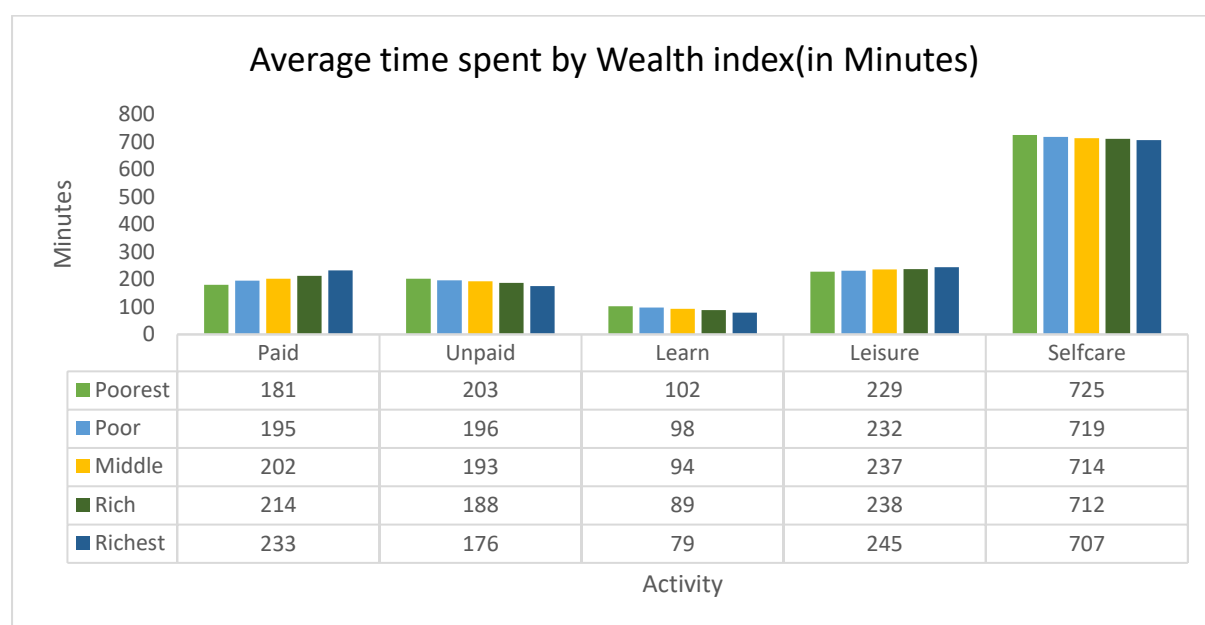


Figure 3.2.5 Average time spent by Education

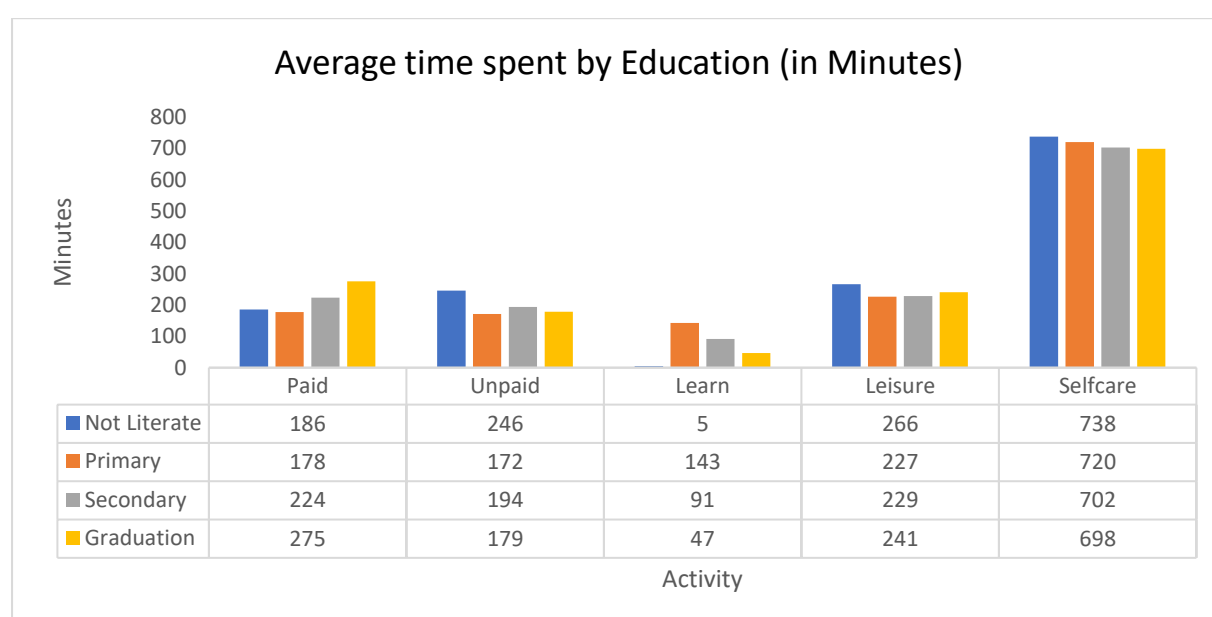


Table 3.2.1 Average time spent by Person age-wise in Different Activities (in Minutes)

Paid Activity

<i>Age Group</i>	Employed	Unemployed	NILF
21–23	468	28	8
24–25	467	29	6
26–27	468	29	6

28–30	466	31	6
Unpaid Activity			
Age Group	Employed	Unemployed	NILF
21–23	79	108	320
24–25	94	113	456
26–27	103	122	503
28–30	115	147	518
Learning Activity			
Age Group	Employed	Unemployed	NILF
21–23	5	173	164
24–25	4	195	50
26–27	3	199	19
28–30	1	164	6

Table 3.2.1 Time use patterns vary across both age groups and employment status. Paid activity remains consistently high for employed individuals across all age groups, with very little variation, while it is minimal for unemployed and NILF individuals. In contrast, unpaid activity shows a clear increasing trend with age, particularly for individuals not in the labour force, indicating a growing involvement in domestic responsibilities. Unemployed individuals also report higher unpaid work compared to the employed. Learning activity is concentrated among younger and unemployed individuals, but declines sharply with age across all groups. Employed individuals consistently spend negligible time on learning. Overall, the results suggest that employment status is the primary determinant of paid activity, while age plays a more important role in shaping unpaid and learning activities.

3.3 Leisure and Self-Care Time Use Patterns among Women

Leisure and self-care are essential for physical and mental well-being, yet women often have limited time for these activities due to work, household responsibilities, and caregiving roles. Time poverty is defined as having insufficient discretionary time after meeting work and domestic obligations is particularly pronounced among working women, restricting their opportunities for rest, recreation, and personal care. Examining these patterns helps understand inequalities in daily routines and the factors influencing women's allocation of time.

Figure 3.3.1 Leisure and self-care time vary across employment status among women. Employed women report the lowest time for both leisure and self-care, reflecting the combined burden of paid work and domestic responsibilities. Unemployed women and those not in the labour force show higher levels of leisure and self-care time. This indicates that participation in paid employment does not reduce domestic responsibilities, but instead compresses women's available personal time.

Figure 3.3.2 Differences in time use are evident across marital status. Widowed women show relatively higher levels of both leisure and self-care time, while currently married women report lower self-care time, suggesting greater household and caregiving responsibilities. Never married and divorced women display moderate and relatively similar patterns. Overall, marital status influences the extent of personal time available to women.

Figure 3.3.3 Leisure and self-care time remain relatively stable across categories of number of children. Only minor variations are observed, indicating that additional children do not substantially alter overall time allocation. This suggests that women's daily schedules are already largely structured around caregiving responsibilities, with limited scope for further adjustment as family size increases.

Figure 3.3.1 Average time spent by women across employment status

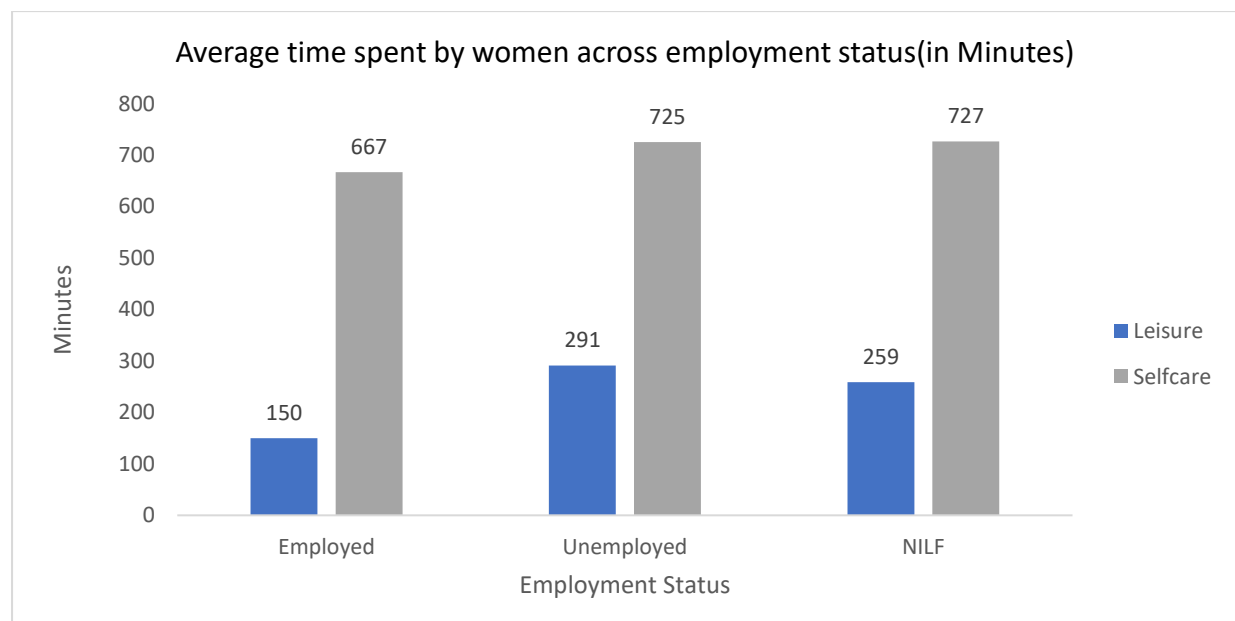


Figure 3.3.2 Average time spent by women across marital status

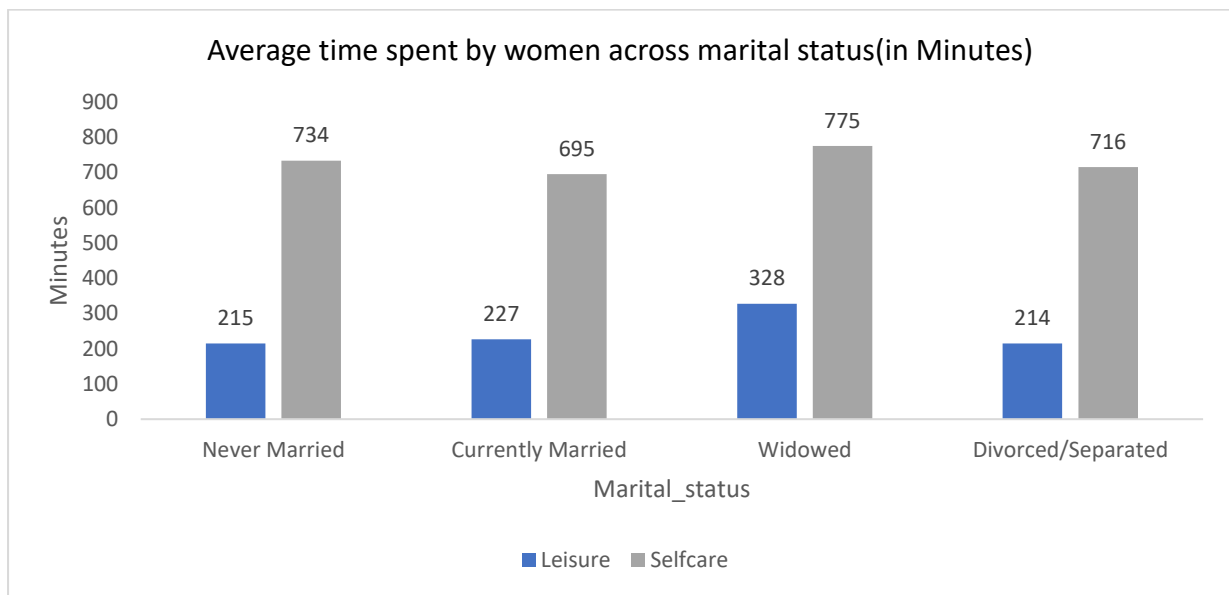


Figure 3.3.3 Average time spent by Married Women by Number of children

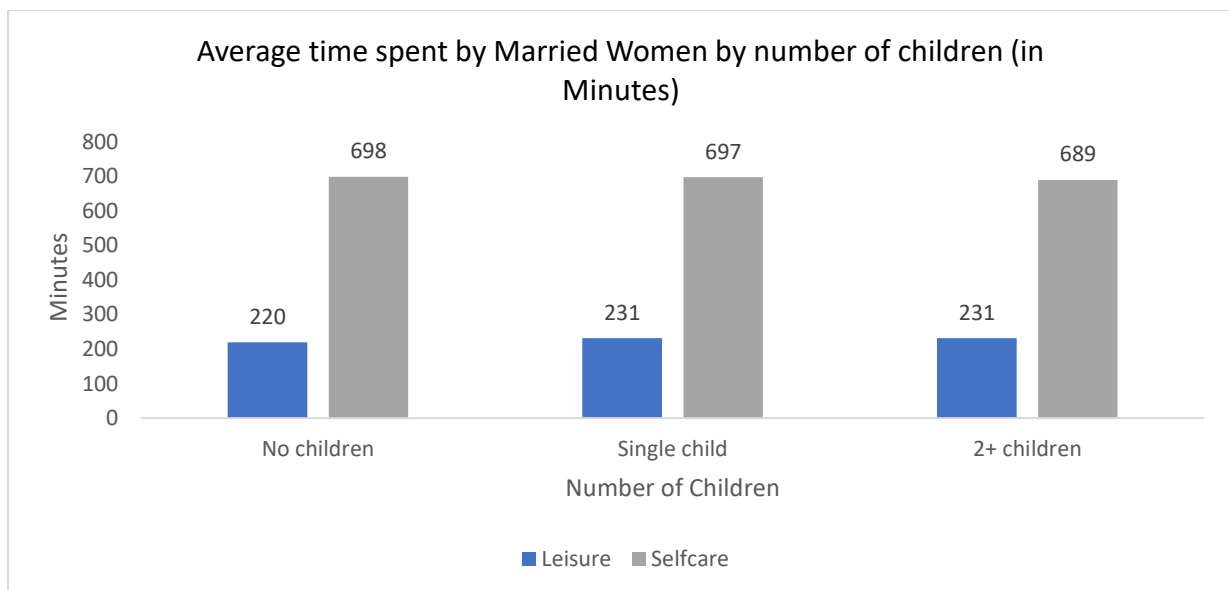


Table 3.3.1 Leisure Time of Married Women by Number of Children and Employment Status (in Minutes)

<i>Number of Children</i>	<i>Employed</i>	<i>Unemployed</i>	<i>NILF</i>
<i>No children</i>	139	234	242
<i>Single child</i>	146	227	259
<i>2+ children</i>	145	261	264

Table 3.3.1 Leisure time among married women varies primarily by employment status, with employed women consistently reporting the lowest leisure time across all categories of number of children. Unemployed women and those not in the labour force show higher levels of leisure time. Variations across number of children are relatively small, indicating that family size does not substantially alter leisure patterns. Overall, employment status appears to be the key factor influencing leisure time among married women.

Table 3.3.2 Average Time Spent in Leisure and Self-Care by Sector and Employment Status among women

Leisure Activity

<i>Sector</i>	Employed	Unemployed	NILF
<i>Rural</i>	149	293	247
<i>Urban</i>	151	290	278

Selfcare Activity

<i>Sector</i>	Employed	Unemployed	NILF
<i>Rural</i>	671	727	727
<i>Urban</i>	658	723	727

Table 3.3.2 Leisure time shows modest variation across sectors, with urban and rural women displaying similar patterns within each employment category. Unemployed individuals report the highest leisure time, followed by those not in the labour force, while employed individuals have the lowest leisure time in both sectors. Self-care time remains relatively stable across sectors, with only slight differences between rural and urban groups. Unemployed women and those not in the labour force report higher self-care time compared to the employed. Overall, the findings suggest that employment status plays a more significant role than sector in shaping both leisure and self-care time use.

3.3.1. The Ordinary Least Square Regression model

The Ordinary Least Squares (OLS) regression model is a widely used statistical method for estimating the relationship between a dependent variable and one or more independent variables. In this study, OLS was applied to examine the factors influencing women's time allocation to leisure, self-care, and the combined total of leisure and self-care. By modelling

these outcomes, the analysis quantifies how socio-economic characteristics, work status, and other relevant factors affect the amount of time women spend on personal well-being activities, providing a clear understanding of time use patterns.

Table 3.3.1.1 Determinants of Self-Care Time among Women

Model summary: N = 200,267, F (27, 9888) = 677.11 | Prob > F = 0.0000, R² = 0.1645

<i>Variable</i>	<i>Coefficient</i>	<i>Std. Err.</i>	<i>t-value</i>	<i>p-value</i>	<i>95% CI</i>
<i>Age Category</i>					
<i>Teenagers (13–19)</i>	-31.716	1.237	-25.64	0.000	[-34.141, -29.292]
<i>Young Adults (20–29)</i>	-24.092	1.805	-13.35	0.000	[-27.630, -20.554]
<i>Adults (30–59)</i>	-21.648	1.930	-11.22	0.000	[-25.431, -17.865]
<i>Elderly (60+)</i>	38.913	2.104	18.50	0.000	[34.789, 43.037]
<i>Marital Status</i>					
<i>Currently Married</i>	-37.620	1.501	-25.07	0.000	[-40.561, -34.678]
<i>Widowed</i>	8.102	1.788	4.53	0.000	[4.597, 11.607]
<i>Divorced/Separated</i>	-1.639	3.870	-0.42	0.672	[-9.226, 5.948]
<i>Education</i>					
<i>Primary</i>	-12.143	0.952	-12.76	0.000	[-14.008, -10.277]
<i>Secondary</i>	-18.400	1.178	-15.62	0.000	[-20.709, -16.091]
<i>Graduation</i>	-16.498	1.508	-10.94	0.000	[-19.455, -13.542]
<i>Household Size</i>					
<i>Medium (4–5)</i>	-1.002	0.853	-1.17	0.240	[-2.674, 0.670]
<i>Large (6+)</i>	5.376	1.204	4.47	0.000	[3.017, 7.736]
<i>Sector</i>					
<i>Urban</i>	4.404	1.473	2.99	0.003	[1.517, 7.292]

<i>Wealth Index</i>					
<i>Poor</i>	-2.093	1.298	-1.61	0.107	[-4.638, 0.451]
<i>Middle</i>	-6.009	1.481	-4.05	0.000	[-8.913, -3.104]
<i>Rich</i>	-6.959	1.656	-4.20	0.000	[-10.204, -3.713]
<i>Richest</i>	-7.060	1.930	-3.66	0.000	[-10.844, -3.277]
<i>No. of Children</i>					
<i>Single Child</i>	1.931	0.829	2.33	0.020	[0.306, 3.556]
<i>2+ Children</i>	-3.443	1.015	-3.39	0.001	[-5.432, -1.455]
<i>Religion</i>					
<i>Islam</i>	-8.676	1.822	-4.76	0.000	[-12.248, -5.105]
<i>Christianity</i>	3.826	2.548	1.50	0.133	[-1.168, 8.819]
<i>Others</i>	5.262	2.845	1.85	0.064	[-0.314, 10.838]
<i>Social Group</i>					
<i>SC</i>	-2.141	2.038	-1.05	0.293	[-6.137, 1.854]
<i>OBC</i>	-4.672	1.984	-2.36	0.019	[-8.561, -0.784]
<i>Others</i>	2.051	2.145	0.96	0.339	[-2.154, 6.257]
<i>Employment Status</i>					
<i>Unemployed</i>	54.776	3.442	15.91	0.000	[48.029, 61.523]
<i>NILF</i>	53.999	1.009	53.52	0.000	[52.022, 55.977]
<i>Constant</i>	724.689	2.666	271.79	0.000	[719.462, 729.915]

Table 3.3.1.1 The regression results indicate that the model is statistically significant ($F = 677.11$, $p < 0.001$), suggesting that the selected socio-economic and demographic variables jointly influence self-care time among women. The R^2 value of 0.1645 implies that approximately 16.45% of the variation in self-care time is explained by the model. Age shows

a strong association, with younger women spending less time on self-care, while elderly women spend significantly more. Marital status is also important, as currently married women report lower self-care time, whereas widowed women report higher levels. Education is negatively associated with self-care time, indicating that more educated women tend to spend less time on personal care. Household size and sector exhibit modest effects, with larger households and urban residence associated with slightly higher self-care time. Wealth shows a negative relationship for middle and higher categories, while the number of children has a limited but significant impact, with larger family size reducing self-care time. Employment status emerges as a key determinant, with unemployed women and those not in the labour force spending significantly more time on self-care compared to employed women. Overall, the findings suggest that self-care time among women is primarily shaped by age, marital status, and employment status, with other factors playing a relatively smaller role.

Table 3.3.1.2 Determinants of Leisure Time among Women

Model Summary: N = 200,267, F (27, 9888) = 921.41 | Prob > F = 0.0000, R² = 0.2464

<i>Variable</i>	<i>Coefficient</i>	<i>Std. Err.</i>	<i>t-value</i>	<i>p-value</i>	<i>95% CI</i>
<i>Age Category</i>					
<i>Teenagers (13–19)</i>	-0.551	1.693	-0.33	0.745	[-3.869, 2.768]
<i>Young Adults (20–29)</i>	52.324	2.376	22.03	0.000	[47.671, 56.984]
<i>Adults (30–59)</i>	88.035	2.515	35.01	0.000	[83.106, 92.964]
<i>Elderly (60+)</i>	180.072	2.805	64.19	0.000	[174.574, 185.571]
<i>Marital Status</i>					
<i>Currently Married</i>	-41.092	1.866	-22.02	0.000	[-44.749, -37.435]
<i>Widowed</i>	6.121	2.273	2.69	0.007	[1.666, 10.576]
<i>Divorced/Separated</i>	-23.730	4.522	-5.25	0.000	[-32.594, -14.866]
<i>Education</i>					
<i>Primary</i>	-8.777	1.211	-7.25	0.000	[-11.151, -6.404]
<i>Secondary</i>	-10.461	1.404	-7.45	0.000	[-13.214, -7.708]

<i>Graduation</i>	-7.428	1.774	-4.19	0.000	[-10.906, -3.951]
<i>Household Size</i>					
<i>Medium (4–5)</i>	0.232	1.055	0.22	0.826	[-1.835, 2.300]
<i>Large (6+)</i>	14.534	1.492	9.74	0.000	[11.609, 17.459]
<i>Sector</i>					
<i>Urban</i>	13.089	1.678	7.80	0.000	[9.799, 16.378]
<i>Wealth Index</i>					
<i>Poor</i>	6.148	1.548	3.97	0.000	[3.113, 9.183]
<i>Middle</i>	9.267	1.714	5.41	0.000	[5.908, 12.626]
<i>Rich</i>	10.643	1.911	5.57	0.000	[6.897, 14.390]
<i>Richest</i>	17.663	2.183	8.09	0.000	[13.385, 21.941]
<i>No. of Children</i>					
<i>Single Child</i>	12.113	1.042	11.63	0.000	[10.071, 14.155]
<i>2+ Children</i>	15.522	1.224	12.68	0.000	[13.122, 17.922]
<i>Religion</i>					
<i>Islam</i>	16.498	2.101	7.85	0.000	[12.379, 20.617]
<i>Christianity</i>	-2.758	2.710	-1.02	0.309	[-8.069, 2.554]
<i>Others</i>	0.513	3.354	0.15	0.879	[-6.061, 7.086]
<i>Social Group</i>					
<i>SC</i>	15.964	2.274	7.02	0.000	[11.507, 20.422]
<i>OBC</i>	15.409	2.154	7.15	0.000	[11.187, 19.631]
<i>Others</i>	20.697	2.299	9.00	0.000	[16.190, 25.205]
<i>Employment Status</i>					
<i>Unemployed</i>	143.598	4.666	30.78	0.000	[134.452, 152.745]

<i>NILF</i>	117.562	1.215	96.78	0.000	[115.181, 119.943]
<i>Constant</i>	54.964	3.133	17.54	0.000	[48.821, 61.106]

Table 3.3.1.2 The regression model is statistically significant ($F = 921.41$, $p < 0.001$), indicating that the explanatory variables jointly influence leisure time among women. The R^2 value of 0.2464 suggests that approximately 24.64% of the variation in leisure time is explained by the model. Age shows a strong positive association, with leisure time increasing significantly across age groups, particularly among elderly women. Marital status is negatively associated with leisure time for currently married and divorced/separated women, while widowed women exhibit slightly higher leisure time. Education is negatively related to leisure time, indicating that more educated women tend to have less leisure. Household size and sector show positive effects, with women in larger households and urban areas reporting higher leisure time. Wealth is positively associated with leisure, with higher wealth groups enjoying greater leisure time. Similarly, the number of children increases leisure time, suggesting possible adjustments in daily routines. Religious and social group differences are present, with some groups reporting higher leisure time, though not all are statistically significant. Employment status emerges as a key determinant, with unemployed women and those not in the labour force having substantially higher leisure time compared to employed women. Overall, leisure time among women is primarily influenced by age, employment status, and socio-economic characteristics, with higher leisure observed among older, non-working, and economically better-off individuals.

Table 3.3.1.3 Determinants of Total Time (Leisure and Self-Care) among Women

Model Summary: $N = 200,267$, $F(27, 9888) = 1855.22$ | $\text{Prob} > F = 0.0000$, $R^2 = 0.3551$

<i>Variable</i>	<i>Coefficient</i>	<i>Std. Err.</i>	<i>t-value</i>	<i>p-value</i>	<i>95% CI</i>
<i>Age Category</i>					
<i>Teenagers (13–19)</i>	-32.267	2.024	-15.94	0.000	[-36.234, -28.299]
<i>Young Adults (20–29)</i>	28.359	2.896	9.75	0.000	[22.559, 33.913]
<i>Adults (30–59)</i>	66.387	3.034	21.88	0.000	[60.441, 72.334]
<i>Elderly (60+)</i>	218.985	3.302	66.32	0.000	[212.513, 225.458]

<i>Marital Status</i>					
<i>Currently Married</i>	-78.712	2.414	-32.60	0.000	[-83.445, -73.979]
<i>Widowed</i>	14.223	2.856	4.98	0.000	[8.624, 19.821]
<i>Divorced/Separated</i>	-25.369	5.542	-4.58	0.000	[-36.233, -14.505]
<i>Education</i>					
<i>Primary</i>	-20.920	1.359	-15.40	0.000	[-23.583, -18.257]
<i>Secondary</i>	-28.861	1.593	-18.12	0.000	[-31.983, -25.739]
<i>Graduation</i>	-23.927	2.054	-11.65	0.000	[-27.954, -19.900]
<i>Household Size</i>					
<i>Medium (4–5)</i>	-0.769	1.210	-0.64	0.525	[-3.142, 1.603]
<i>Large (6+)</i>	19.910	1.637	12.16	0.000	[16.701, 23.120]
<i>Sector</i>					
<i>Urban</i>	17.493	1.686	10.37	0.000	[14.188, 20.798]
<i>Wealth Index</i>					
<i>Poor</i>	4.055	1.628	2.49	0.013	[0.863, 7.247]
<i>Middle</i>	3.258	1.808	1.80	0.072	[-0.285, 6.802]
<i>Rich</i>	3.685	2.014	1.83	0.067	[-0.263, 7.633]
<i>Richest</i>	10.603	2.288	4.64	0.000	[6.119, 15.087]
<i>No. of Children</i>					
<i>Single Child</i>	14.044	1.183	11.87	0.000	[11.725, 16.362]
<i>2+ Children</i>	12.078	1.406	8.59	0.000	[9.323, 14.834]
<i>Religion</i>					
<i>Islam</i>	7.821	2.147	3.64	0.000	[3.612, 12.030]
<i>Christianity</i>	1.068	3.239	0.33	0.742	[-5.280, 7.416]

<i>Others</i>	5.774	3.734	1.55	0.122	[-1.545, 13.094]
<i>Social Group</i>					
<i>SC</i>	13.823	2.477	5.58	0.000	[8.968, 18.677]
<i>OBC</i>	10.737	2.364	4.54	0.000	[6.102, 15.371]
<i>Others</i>	22.748	2.543	8.95	0.000	[17.764, 27.732]
<i>Employment Status</i>					
<i>Unemployed</i>	198.375	6.036	32.87	0.000	[186.543, 210.206]
<i>NILF</i>	171.561	1.474	116.38	0.000	[168.672, 174.451]
<i>Constant</i>	779.652	3.591	217.14	0.000	[772.614, 786.691]

Table 3.3.1.3 The regression model is statistically significant ($F = 1855.22$, $p < 0.001$), indicating that the explanatory variables jointly influence total time spent on leisure and self-care among women. The R^2 value of 0.3551 suggests that approximately 35.51% of the variation in total time is explained by the model, indicating a relatively stronger explanatory power compared to individual activity models. Age shows a strong positive association, with total time increasing significantly across age groups, particularly among elderly women. Marital status is negatively associated for currently married and divorced/separated women, while widowed women report higher total time. Education is negatively related, indicating that more educated women tend to have lower combined leisure and self-care time. Household size and sector exhibit positive effects, with larger households and urban residence associated with higher total time. Wealth shows a modest positive association, particularly for the richest group. Similarly, the number of children increases total time, suggesting adjustments in daily time allocation. Religious and social group differences are present, with some groups reporting higher total time, though not all coefficients are statistically significant. Employment status emerges as the most influential factor, with unemployed women and those not in the labour force reporting substantially higher total time compared to employed women. Overall, the findings indicate that total non-work time among women is primarily shaped by age, employment status, and marital status, with socio-economic characteristics playing a secondary role.

3.4 Childcare Time Use Pattern

Childcare forms an essential component of daily time use, as it requires substantial and often inflexible time commitments, thereby influencing how individuals allocate time across work, leisure, and personal activities. Both younger adults and elderly caregivers (above 60 years) contribute to childcare, highlighting intergenerational responsibilities within households. Examining childcare time use patterns provides insights into the distribution of caregiving duties, the burden of unpaid work, and variations across different demographic and socio-economic groups.

Figure 3.4.1 The age distribution of childcare participants is concentrated primarily in the 20–40 years range, with a clear peak in the late 20s to early 30s. This suggests that individuals in their prime parenting years are most actively engaged in childcare activities.

Participation declines progressively beyond age 40, and very few individuals in older age groups are involved. The distribution is positively skewed, indicating a longer tail toward higher ages. Overall, the pattern highlights a strong association between age and childcare participation, with engagement highest during the typical child-rearing period.

Figure 3.4.1 Distribution of Childcare Participant Across Age

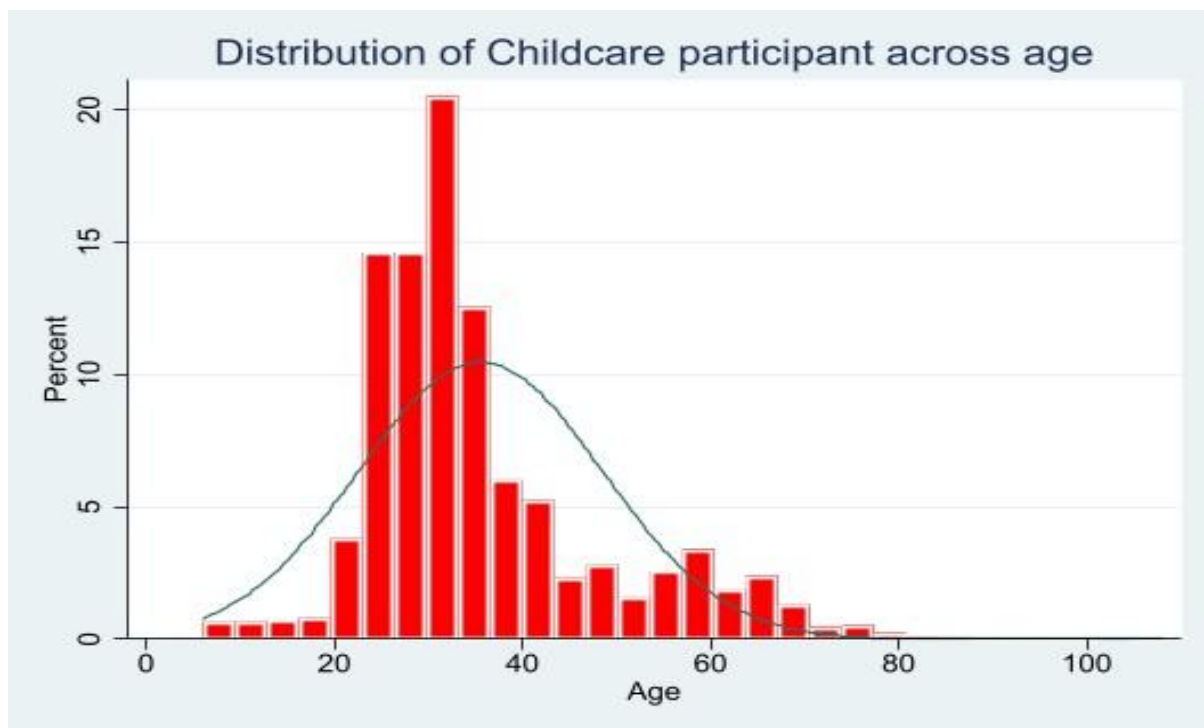


Table 3.4.1 Average Time Spent in Childcare Activities (in Minutes)

Per Person Mean: 30 Minutes

<i>Variable</i>	<i>Category</i>	<i>Mean</i>
<i>Gender</i>	Male	13
	Female	46
<i>Age Category</i>	Children (6–12)	3
	Teenagers (13–19)	4
	Young Adults (20–29)	58
	Adults (30–59)	31
	Elderly (60+)	17
<i>Marital Status</i>	Never Married	2
	Currently Married	43
	Widowed	24
	Divorced/Separated	24
<i>Employment Status</i>	Employed	19
	Unemployed	12
	NILF	39
<i>Wealth Index</i>	Poorest	41
	Poor	34
	Middle	30
	Rich	25
	Richest	18

Table 3.4.1 presents the distribution of time spent on childcare activities across different socio-demographic groups. The overall per person mean indicates a moderate level of engagement in childcare, with considerable variation across categories. A pronounced gender disparity is observed, with females allocating substantially more time to childcare than males, reflecting

the persistence of traditional gender roles in the division of unpaid care work. Age-wise patterns show that individuals in the young adult and adult age groups contribute the most to childcare, indicating their primary responsibility in caregiving within households. In contrast, children and teenagers exhibit minimal involvement, while the elderly demonstrate a moderate level of participation, suggesting some degree of intergenerational support. Marital status further influences childcare time allocation. Currently married individuals record the highest levels of engagement, underscoring their central role in family-based caregiving. Those who are widowed or divorced/separated also contribute a notable amount of time, whereas never married individuals show minimal involvement, likely due to lower caregiving responsibilities.

Variation across employment status highlights the role of time availability. Individuals outside the labour force allocate the most time to childcare, followed by those who are employed, while the unemployed report the lowest levels of engagement. This suggests that competing demands of paid work and labour market attachment shape the extent of participation in childcare activities. An inverse gradient is evident across wealth groups, with individuals from lower economic strata devoting more time to childcare compared to their wealthier counterparts. This pattern may reflect differences in access to market-based childcare services, leading to a greater reliance on household labour among poorer groups. Overall, the findings reveal significant socio-economic and demographic differentials in childcare time allocation, emphasizing the unequal distribution of unpaid care responsibilities within households.

3.5. Discussion

This chapter highlights the unequal distribution of time across activities and population groups. The analysis shows a substantial concentration of time in unpaid activity, with clear variations across socio-demographic and economic characteristics. The examination of leisure and self-care among women indicates limited time availability, reflecting the burden of unpaid responsibilities and the presence of time poverty. The OLS results further establish that age, marital status, employment, and economic position significantly influence these patterns. The analysis of childcare time use across all individuals reveals notable disparities, particularly along gender and wealth dimensions, with women and economically disadvantaged groups bearing a greater share of caregiving responsibilities. These patterns demonstrate that time allocation is shaped by underlying social and economic structures rather than being evenly distributed. The empirical findings presented in this chapter provide a basis for the subsequent analysis, particularly in assessing the reliability and precision of the estimates through the examination of sampling error in the following chapter.

CHAPTER 4

ASSESSMENT OF SAMPLING ERRORS USING DESIGN- BASED VARIANCE ESTIMATION METHODS

4.1. Introduction:

Design-based variance estimation methods are used to assess sampling error in survey data by relying on the sampling design. These methods account for complex survey features such as stratification, clustering, and unequal selection probabilities, ensuring accurate estimation of variance. Common techniques include Taylor Series Linearization and replication methods like Jackknife and Bootstrap. Design-based approaches provide reliable estimates of standard errors, relative standard errors (RSE), and confidence intervals, making them essential for evaluating the precision and reliability of survey estimates.

4.2. Results of Sampling Error for Total Unpaid Time

Table (4.2.1 – 4.2.7) The analysis of sampling errors for total unpaid time provides a clear understanding of the reliability and precision of the survey estimates. At the aggregate level, total unpaid time is highly robust, with Taylor Linearization, Jackknife, and Bootstrap methods all producing identical means of 191.82 minutes, narrow confidence intervals, and low relative standard errors of 0.33 percent. This indicates minimal sampling error and high precision in estimating total unpaid time for the overall population. The consistency across methods also suggests that the complex survey design, including stratification and clustering, is well accounted for in variance estimation.

Subgroup analyses of total unpaid time reveal important patterns in sampling variability. Larger population groups, including males, females, rural and urban residents, and main educational categories, display low standard errors, narrow confidence intervals, and small relative standard errors, reflecting sufficient sample sizes and low heterogeneity. Smaller or less-represented groups, particularly the unemployed, divorced or separated, and never-married, exhibit higher standard errors, wider confidence intervals, and elevated relative standard errors. For example, the divorced or separated category shows a relative standard error of 2.43 to 2.51 percent and a considerably wider 95 percent confidence interval, indicating increased

uncertainty due to smaller sample sizes and higher variability in total unpaid time within this subgroup. This highlights the need to consider sampling variability when interpreting subgroup estimates of total unpaid time, as reliability is inherently lower for smaller or more heterogeneous groups.

Design effects, DEFF, further clarify the impact of survey design on variability in total unpaid time. DEFF values vary across subgroups, with higher values for males and employed individuals, suggesting that clustering and unequal weighting inflate variance in these groups. Bootstrap methods generally produce slightly higher DEFF values than Taylor Linearization and Jackknife because they capture additional variability through resampling, yet all methods yield broadly consistent estimates, confirming the robustness of variance estimation.

Overall, the low sampling variability for most subgroups, combined with narrow confidence intervals and low relative standard errors, provides strong evidence that the estimates of total unpaid time are reliable and precise. The examination of subgroup variability and design effects demonstrates how survey design, sample size, and population heterogeneity interact to influence the precision of total unpaid time estimates, offering a comprehensive understanding of the strengths and limitations of the data.

Table 4.2.1 Sampling Error Measures of Total Unpaid Time

<i>Method</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>CI</i>	<i>RSE</i>
<i>Taylor Linearization</i>					
<i>Unpaid time</i>	191.82	0.63	3.26	190.59 - 193.05	0.33
<i>Jackknife Method</i>					
<i>Unpaid time</i>	191.82	0.63	3.26	190.59 - 193.05	0.33
<i>Bootstrap Method</i>					
<i>Unpaid time</i>	191.82	0.63	3.33	190.58 - 193.06	0.33

Table 4.2.2 Sampling Error Measures of Total Unpaid Time by Gender

<i>Method</i>	<i>Gender</i>	<i>Mean</i> <i>(minutes)</i>	<i>Std. Error</i>	<i>95% CI</i>	<i>DEFF</i>	<i>RSE</i> <i>(%)</i>
<i>Taylor Linearization</i>	Male	54.11	0.53	53.07–55.15	5.88	0.98

<i>Jackknife</i>	Female	327.25	0.96	325.36–329.14	3.82	0.29
	Male	54.11	0.54	53.06–55.15	5.98	0.99
<i>Bootstrap</i>	Female	327.25	0.97	325.37–329.14	3.81	0.29
	Male	54.11	0.55	53.05–55.16	6.00	1.02
	Female	327.25	0.98	325.35–329.14	3.80	0.30

Table 4.2.3 Sampling Error Measures of Total Unpaid Time by Sector

<i>Method</i>	<i>Sector</i>	<i>Mean (minutes)</i>	<i>Std. Error</i>	<i>95% CI</i>	<i>DEFF</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Rural	201.25	0.82	199.64–202.86	3.72	0.41
	Urban	170.73	0.83	169.11–172.36	1.97	0.49
<i>Jackknife</i>	Rural	201.25	0.82	199.64–202.86	3.72	0.41
	Urban	170.73	0.83	169.11–172.36	1.97	0.49
<i>Bootstrap</i>	Rural	201.25	0.81	199.67–202.83	3.26	0.40
	Urban	170.73	0.84	169.08–172.39	2.45	0.49

Table 4.2.4 Sampling Error Measures of Total Unpaid Time by Employment Status

<i>Method</i>	<i>Employment Status</i>	<i>Mean (minutes)</i>	<i>Std. Error</i>	<i>95% CI</i>	<i>DEFF</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Employed	115.52	0.87	113.80–117.23	5.90	0.76
	Unemployed	114.61	2.62	109.48–119.74	1.90	2.28
	NILF	259.58	0.99	257.63–261.53	3.47	0.38
<i>Jackknife</i>	Employed	115.52	0.87	113.80–117.23	5.90	0.76

<i>Bootstrap</i>	Unemployed	114.61	2.62	109.47–119.74	1.90	2.29
	NILF	259.58	0.99	257.63–261.53	3.47	0.38
	Employed	115.52	0.90	113.75–117.28	6.24	0.78
	Unemployed	114.61	2.64	109.43–119.78	2.07	2.30
	NILF	259.58	1.02	257.58–261.58	3.67	0.39

Table 4.2.5 Sampling Error Measures of Total Unpaid Time by Education

<i>Method</i>	<i>Education Level</i>	<i>Mean (minutes)</i>	<i>Std. Error</i>	<i>95% CI</i>	<i>DEFF</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Not Literate	248.73	1.39	246.01–251.45	3.18	0.56
	Primary	170.34	0.84	168.70–171.99	2.53	0.49
	Secondary	191.91	0.94	190.07–193.76	1.86	0.49
	Graduation	178.62	1.26	176.15–181.08	1.71	0.70
<i>Jackknife</i>	Not Literate	248.73	1.39	246.01–251.46	2.93	0.56
	Primary	170.34	0.84	168.70–171.98	2.50	0.49
	Secondary	191.91	0.94	190.07–193.76	1.93	0.49
	Graduation	178.62	1.23	176.20–181.04	1.81	0.69
<i>Bootstrap</i>	Not Literate	248.73	1.39	246.01–251.46	2.93	0.56
	Primary	170.34	0.84	168.70–171.98	2.50	0.49
	Secondary	191.91	0.94	190.07–193.76	1.93	0.49
	Graduation	178.62	1.23	176.20–181.04	1.81	0.69

Table 4.2.6 Sampling Error Measures of Total Unpaid Time by Wealth Index

<i>Method</i>	<i>Group / Quintile</i>	<i>Mean (minutes)</i>	<i>Std. Error</i>	<i>95% CI</i>	<i>DEFF</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Poorest	205.05	1.30	202.50–207.60	2.84	0.63
	Poor	197.42	1.06	195.20–199.49	1.83	0.54
	Middle	193.29	0.97	191.38–195.35	1.53	0.50
	Rich	186.74	0.93	184.92–188.55	1.39	0.50
	Richest	173.11	0.97	171.20–175.02	1.76	0.56
<i>Jackknife</i>	Poorest	205.05	1.30	202.50–207.60	2.66	0.63
	Poor	197.42	1.09	195.29–199.56	1.88	0.55
	Middle	193.29	0.96	191.42–195.17	1.49	0.49
	Rich	186.74	0.92	184.93–188.55	1.44	0.49
	Richest	173.11	0.98	171.18–175.04	1.90	0.57
<i>Bootstrap</i>	Poorest	205.05	1.30	202.50–207.60	2.66	0.63
	Poor	197.42	1.09	195.29–199.56	1.88	0.55
	Middle	193.29	0.96	191.42–195.17	1.49	0.49
	Rich	186.74	0.92	184.93–188.55	1.44	0.49
	Richest	173.11	0.98	171.18–175.04	1.90	0.57

Table 4.2.7 Sampling Error Measures of Total Unpaid Time by Marital Status

<i>Method</i>	<i>Marital Status</i>	<i>Mean (minutes)</i>	<i>Std. Error</i>	<i>95% CI</i>	<i>DEFF</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Never Married	50.36	0.52	49.34–51.37	2.64	1.03

<i>Jackknife</i>	Currently Married	256.91	0.79	255.36–258.47	2.94	0.31
	Widowed	200.05	1.26	197.58–202.52	1.64	0.63
	Divorced/Separated	185.88	4.52	177.02–194.74	1.33	2.43
	Never Married	50.36	0.53	49.31–51.40	2.80	1.06
<i>Bootstrap</i>	Currently Married	256.91	0.81	255.33–258.49	3.02	0.31
	Widowed	200.05	1.26	197.57–202.53	1.66	0.63
	Divorced/Separated	185.88	4.66	176.75–195.01	1.55	2.51
	Never Married	50.36	0.53	49.31–51.40	2.80	1.06
	Currently Married	256.91	0.81	255.33–258.49	3.02	0.31
	Widowed	200.05	1.26	197.57–202.53	1.66	0.63
	Divorced/Separated	185.88	4.66	176.75–195.01	1.55	2.51
	Never Married	50.36	0.53	49.31–51.40	2.80	1.06

4.3. Results of Sampling Error for Time Poverty Rate

Table (4.3.1 – 4.3.5) The analysis of sampling errors for the time poverty rate demonstrates that the survey estimates are generally precise and reliable. At the overall level, the meantime poverty rate is 0.26 across Taylor Linearization, Jackknife, and Bootstrap methods, with narrow confidence intervals (0.26 to 0.27) and low relative standard errors (1.47–1.48 percent). This indicates minimal sampling error and high precision in estimating the time poverty rate for the total population. The consistency across variance estimation methods further confirms that the survey design, including stratification and clustering, has been effectively accounted for.

Subgroup analyses reveal variations in sampling variability across population groups. Larger and well-represented groups, such as rural and urban residents, and those in the main education or wealth categories, show relatively low standard errors, narrow confidence intervals, and moderate relative standard errors, reflecting sufficient sample sizes and low heterogeneity. In contrast, smaller or less-represented groups, particularly the never-married and divorced/separated categories, display higher standard errors, wider confidence intervals, and elevated relative standard errors, with the divorced/separated group reaching an RSE of 7.2–

7.36 percent. This highlights that estimates for these subgroups are less precise due to smaller sample sizes and higher variability in time poverty within these populations.

Design effects provide further insight into how the complex survey design affects the variability of time poverty estimates. DEFF values vary across subgroups, with higher values observed for currently married individuals and rural residents, indicating that clustering and unequal weighting inflate variance in these groups. Bootstrap methods generally produce slightly higher DEFF values compared to Taylor Linearization and Jackknife, reflecting additional variability captured through resampling, while maintaining consistency with other methods. Overall, the low sampling variability for the majority of subgroups, combined with narrow confidence intervals and moderate relative standard errors, indicates that the survey estimates of the time poverty rate are robust and reliable. The detailed examination of subgroup variability and design effects emphasizes the importance of accounting for sampling error in interpreting time poverty estimates, particularly for smaller or more heterogeneous population groups.

Table 4.3.1 Sampling Error Measures of Time Poverty Rate

<i>Method</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>CI</i>	<i>RSE</i>
<i>Taylor Linearization</i>					
<i>Time poor</i>	0.26	0.004	3.32	0.26 - 0.27	1.47
<i>Jackknife Method</i>					
<i>Time poor</i>	0.26	0.004	3.33	0.26 - 0.27	1.47
<i>Bootstrap Method</i>					
<i>Time poor</i>	0.26	0.004	3.40	0.26 - 0.27	1.48

Table 4.3.2 Sampling Error Measures of Time Poverty Rate by Sector

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Rural	0.26	0.005	3.59	0.26–0.27	1.78
	Urban	0.26	0.007	2.61	0.25–0.27	2.53

<i>Jackknife</i>	Rural	0.26	0.005	3.59	0.26–0.27	1.78
	Urban	0.26	0.007	2.61	0.25–0.27	2.53
<i>Bootstrap</i>	Rural	0.26	0.005	3.33	0.26–0.27	1.77
	Urban	0.26	0.007	3.23	0.25–0.27	2.60

Table 4.3.3 Sampling Error Measures of Time Poverty Rate by Education

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Not Literate	0.26	0.006	2.57	0.25–0.27	2.17
	Primary	0.28	0.005	2.20	0.26–0.29	1.97
	Secondary	0.27	0.007	1.85	0.26–0.29	2.59
	Graduation	0.23	0.008	1.84	0.21–0.24	3.36
<i>Jackknife</i>	Not Literate	0.26	0.006	2.57	0.25–0.27	2.17
	Primary	0.28	0.005	2.20	0.26–0.29	1.98
	Secondary	0.27	0.007	1.86	0.26–0.29	2.59
	Graduation	0.23	0.008	1.84	0.21–0.24	3.36
<i>Bootstrap</i>	Not Literate	0.26	0.006	2.36	0.25–0.27	2.18
	Primary	0.28	0.006	2.30	0.26–0.29	2.01
	Secondary	0.27	0.008	2.35	0.26–0.29	2.79
	Graduation	0.23	0.008	2.06	0.21–0.24	3.36

Table 4.3.4 Sampling Error Measures of Time Poverty Rate by Wealth Index

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Poorest	0.29	0.008	2.57	0.28–0.31	2.61
	Poor	0.27	0.007	2.05	0.25–0.28	2.52

<i>Jackknife</i>	Middle	0.26	0.006	1.87	0.24–0.27	2.53
	Rich	0.26	0.007	2.06	0.25–0.28	2.64
	Richest	0.24	0.007	2.11	0.22–0.25	2.84
	Poorest	0.29	0.008	2.58	0.28–0.31	2.61
	Poor	0.27	0.007	2.05	0.25–0.28	2.53
<i>Bootstrap</i>	Middle	0.26	0.006	1.87	0.24–0.27	2.53
	Rich	0.26	0.007	2.06	0.25–0.28	2.64
	Richest	0.24	0.007	2.12	0.22–0.25	2.84
	Poorest	0.29	0.008	2.68	0.28–0.31	2.70
	Poor	0.27	0.007	2.08	0.25–0.28	2.57
	Middle	0.26	0.007	2.08	0.24–0.27	2.65
	Rich	0.26	0.007	2.22	0.25–0.28	2.71
	Richest	0.24	0.007	2.16	0.22–0.25	2.84

Table 4.3.5 Sampling Error Measures of Time Poverty Rate by Marital Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Never Married	0.15	0.007	1.87	0.13–0.16	5.07
	Currently Married	0.28	0.004	3.12	0.27–0.29	1.57
	Widowed	0.24	0.006	1.45	0.23–0.26	2.64
	Divorced/Separated	0.30	0.022	1.28	0.25–0.34	7.33
<i>Jackknife</i>	Never Married	0.15	0.007	1.88	0.13–0.16	5.08
	Currently Married	0.28	0.004	3.12	0.27–0.29	1.58
	Widowed	0.24	0.006	1.45	0.23–0.26	2.64
	Divorced/Separated	0.30	0.022	1.29	0.25–0.34	7.36

<i>Bootstrap</i>	Never Married	0.15	0.007	2.09	0.13–0.16	5.08
	Currently Married	0.28	0.004	3.07	0.27–0.29	1.58
	Widowed	0.24	0.007	1.58	0.23–0.26	2.76
	Divorced/Separated	0.30	0.021	1.41	0.25–0.34	7.20

4.4. Results of Sampling Error for Childcare Rate

Table (4.4.1 – 4.4.7) The analysis of sampling errors for the childcare rate indicates that the survey estimates are generally precise and reliable. At the overall level, the mean childcare rate is 0.15 across Taylor Linearization, Jackknife, and Bootstrap methods, with narrow confidence intervals (0.15 to 0.16) and low relative standard errors (1.46–1.49 percent). This suggests minimal sampling error and high precision in estimating childcare participation for the total population. The close agreement across variance estimation methods demonstrates that the complex survey design, including stratification and clustering, has been appropriately accounted for.

Subgroup analyses reveal variations in sampling variability across population groups. Larger and well-represented groups, such as males, females, rural and urban residents, and major educational or wealth categories, exhibit relatively low standard errors, narrow confidence intervals, and modest relative standard errors, reflecting sufficient sample sizes and low heterogeneity. In contrast, smaller or less-represented groups, particularly the unemployed, never-married, and divorced or separated categories, show higher standard errors, wider confidence intervals, and elevated relative standard errors. For instance, the unemployed category demonstrates an RSE of 25.86–27.08 percent, while the never-married subgroup shows an RSE of approximately 19.75–19.94 percent, highlighting greater uncertainty in these estimates due to smaller sample sizes and higher variability in childcare responsibilities.

Design effects (DEFF) further illustrate the influence of the survey design on variability. Higher DEFF values are observed for currently married individuals and rural residents, indicating that clustering and unequal weighting contribute to increased variance in these subgroups. Bootstrap methods generally produce slightly higher DEFF values than Taylor Linearization and Jackknife, reflecting the additional variability captured through resampling, while maintaining consistency with the other methods.

Overall, the low sampling variability for most subgroups, combined with narrow confidence intervals and low to moderate relative standard errors, indicates that the childcare rate estimates are robust and reliable. Examining subgroup variability and design effects underscores the importance of accounting for sampling error when interpreting estimates, particularly for smaller or more heterogeneous groups, to ensure a correct understanding of childcare participation patterns across the population.

Table 4.4.1 Sampling Error Measures of Childcare Rate

<i>Method</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>CI</i>	<i>RSE</i>
<i>Taylor Linearization</i>					
Childcare	0.15	0.002	2.04	0.15- 0.16	1.46
<i>Jackknife Method</i>					
Childcare	0.15	0.002	2.04	0.15- 0.16	1.46
<i>Bootstrap Method</i>					
Childcare	0.15	0.002	2.13	0.15- 0.16	1.49

Table 4.4.2 Sampling Error Measures of Childcare Rate by Gender

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Male	0.12	0.003	1.58	0.11–0.12	2.12
	Female	0.19	0.003	1.56	0.18–0.20	1.59
<i>Jackknife</i>	Male	0.12	0.003	1.58	0.11–0.12	2.12
	Female	0.19	0.003	1.56	0.18–0.20	1.59
<i>Bootstrap</i>	Male	0.12	0.002	1.53	0.11–0.12	2.09
	Female	0.19	0.003	1.70	0.18–0.20	1.65

Table 4.4.3 Sampling Error Measures of Childcare Rate by Sector

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Rural	0.16	0.003	2.08	0.15–0.16	1.76
	Urban	0.15	0.004	1.93	0.14–0.16	2.60
<i>Jackknife</i>	Rural	0.16	0.003	2.09	0.15–0.16	1.76
	Urban	0.15	0.004	1.93	0.14–0.16	2.60
<i>Bootstrap</i>	Rural	0.16	0.003	1.91	0.15–0.16	1.77
	Urban	0.15	0.004	2.41	0.14–0.16	2.63

Table 4.4.4 Sampling Error Measures of Childcare Rate by Employment Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Employed	0.13	0.003	1.70	0.12–0.13	2.52
	Unemployed	0.21	0.054	1.16	0.10–0.32	25.86
	NILF	0.17	0.003	1.81	0.16–0.17	1.62
<i>Jackknife</i>	Employed	0.13	0.003	1.70	0.12–0.13	2.52
	Unemployed	0.21	0.056	1.23	0.10–0.32	26.72
	NILF	0.17	0.003	1.81	0.16–0.17	1.62
<i>Bootstrap</i>	Employed	0.13	0.003	1.66	0.12–0.13	2.55
	Unemployed	0.21	0.057	1.40	0.10–0.32	27.08
	NILF	0.17	0.003	1.91	0.16–0.17	1.64

Table 4.4.5 Sampling Error Measures of Childcare Rate by Education

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Not literate	0.17	0.003	1.78	0.17–0.18	1.84

<i>Jackknife</i>	Primary	0.15	0.004	1.70	0.14–0.16	2.44
	Secondary	0.13	0.005	1.34	0.12–0.14	3.80
	Graduation	0.10	0.006	1.30	0.09–0.12	5.81
	Not literate	0.17	0.003	1.78	0.17–0.18	1.84
	Primary	0.15	0.004	1.70	0.14–0.16	2.44
<i>Bootstrap</i>	Secondary	0.13	0.005	1.54	0.12–0.14	3.80
	Graduation	0.10	0.006	1.38	0.09–0.12	5.83
	Not literate	0.17	0.003	1.79	0.17–0.18	1.90
	Primary	0.15	0.004	1.72	0.14–0.16	2.43
	Secondary	0.13	0.005	1.65	0.12–0.14	3.76
	Graduation	0.10	0.006	1.68	0.09–0.12	5.92

Table 4.4.6 Sampling Error Measures of Childcare Rate by Wealth Index

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Poorest	0.24	0.005	1.87	0.23–0.25	2.27
	Poor	0.19	0.005	1.80	0.18–0.20	2.63
	Middle	0.15	0.005	1.84	0.14–0.16	3.17
	Rich	0.11	0.004	1.70	0.10–0.11	3.79
	Richest	0.07	0.004	2.12	0.06–0.07	5.68
<i>Jackknife</i>	Poorest	0.24	0.005	1.87	0.23–0.25	2.27
	Poor	0.19	0.005	1.80	0.18–0.20	2.63
	Middle	0.15	0.005	1.84	0.14–0.16	3.17
	Rich	0.11	0.004	1.70	0.10–0.11	3.79
	Richest	0.07	0.004	2.13	0.06–0.07	5.69

Bootstrap

Poorest	0.24	0.005	1.76	0.23–0.25	2.25
Poor	0.19	0.005	1.84	0.18–0.20	2.71
Middle	0.15	0.005	1.75	0.14–0.16	3.10
Rich	0.11	0.004	1.77	0.10–0.11	3.81
Richest	0.07	0.004	2.44	0.06–0.07	5.88

Table 4.4.7 Sampling Error Measures of Childcare Rate by Marital Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>DEFF</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>Taylor Linearization</i>	Never Married	0.07	0.014	1.21	0.04–0.10	19.88
	Currently Married	0.14	0.003	2.03	0.14–0.15	1.90
	Widowed	0.18	0.004	1.49	0.17–0.19	1.96
	Divorced/Separated	0.13	0.022	1.15	0.09–0.17	17.30
<i>Jackknife</i>	Never Married	0.07	0.014	1.22	0.04–0.10	19.94
	Currently Married	0.14	0.003	2.04	0.14–0.15	1.90
	Widowed	0.18	0.004	1.49	0.17–0.19	1.95
	Divorced/Separated	0.13	0.023	1.16	0.09–0.17	17.41
<i>Bootstrap</i>	Never Married	0.07	0.014	1.24	0.04–0.10	19.75
	Currently Married	0.14	0.003	2.07	0.14–0.15	1.92
	Widowed	0.18	0.004	1.66	0.17–0.19	2.06
	Divorced/Separated	0.13	0.023	1.18	0.09–0.17	17.35

4.5. Discussion

Design-based variance estimation indicates that estimates of total unpaid time, time poverty, and childcare rate are generally precise, with low standard errors, narrow confidence intervals, and small relative standard errors. Taylor Linearization, Jackknife, and Bootstrap methods yield consistent results, reflecting proper accounting for survey design, including stratification, clustering, and weighting. Design effects (DEFF) reveal the impact of complex survey design on variance: higher DEFF values for certain subgroups, such as males, employed, currently married, and rural respondents, indicate that clustering and unequal weighting increase variance in these groups. Smaller or less-represented subgroups, such as the unemployed or divorced/separated, show wider confidence intervals and higher RSEs, demonstrating greater uncertainty. Overall, DEFF highlights how survey design features amplify variability in some subgroups, while estimates for larger, more homogeneous groups remain robust and reliable.

CHAPTER 5

ASSESSMENT OF SAMPLING ERRORS USING MODEL-BASED VARIANCE ESTIMATION METHODS

5.1 Introduction

Model-based variance estimation provides an alternative framework for assessing the precision of survey estimates by deriving variability from a specified statistical model rather than relying solely on the survey design. This approach allows for efficient estimation of parameters while accounting for unequal observation weights and the structure of the data. In this study, Weighted Least Squares (WLS) and Bootstrap WLS methods were applied to estimate variance for key time use indicators. WLS produces unbiased and efficient estimates by incorporating survey weights, whereas Bootstrap WLS leverages repeated resampling to empirically assess variability and construct robust confidence intervals. By complementing design-based methods, model-based variance estimation provides a comprehensive understanding of sampling variability and the stability of estimated indicators.

5.2. Results of Sampling Error for Total Unpaid Time

Table (5.2.1- 5.2.7) The estimation of total unpaid time using Weighted Least Squares (WLS) and Bootstrap WLS demonstrates high precision and reliability. The overall mean unpaid time is 191.82 minutes, with very low standard errors and relative standard errors, and narrow 95% confidence intervals (191.14–192.50 for WLS and 191.06–192.59 for Bootstrap WLS), indicating that the population mean is estimated with high certainty.

Subgroup analysis reveals consistent patterns in unpaid time is significantly higher among females, rural residents, married individuals, and those not in the labour force, while it decreases among males, urban residents, and higher-wealth or more educated groups. Smaller subgroups, such as the unemployed or divorced/separated individuals, show slightly higher variability, reflected in wider confidence intervals, but overall estimates remain reliable.

Both methods provide consistent results, with Bootstrap WLS producing marginally wider confidence intervals due to resampling variability. The low RSEs and narrow confidence intervals across most categories confirm the robustness of the estimates, supporting confident

interpretation of the distribution and determinants of total unpaid time across demographic and socioeconomic groups.

Table 5.2.1 Sampling Error Measures of Total Unpaid Time

<i>Method</i>	<i>Mean</i>	<i>SE</i>	<i>CI</i>	<i>RSE</i>
<i>WLS</i>				
<i>Unpaid time</i>	191.82	0.347	191.14- 192.50	0.18
<i>Bootstrap WLS</i>				
<i>Unpaid time</i>	191.82	0.39	191.06- 192.59	0.2

Table 5.2.2 Sampling Error Measures of Total Unpaid Time by Gender

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Male	54.11	0.385	53.35–54.86	0.71
	Female	327.25	0.382	326.50–328.00	0.12
<i>Bootstrap WLS</i>	Male	54.11	0.239	53.61–54.59	0.44
	Female	327.25	0.585	326.14–328.42	0.18

Table 5.2.3 Sampling Error Measures of Total Unpaid Time by Sector

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Rural	201.25	0.417	200.43–202.07	0.21
	Urban	170.73	0.623	169.51–171.96	0.36
<i>Bootstrap WLS</i>	Rural	201.25	0.497	200.33–202.29	0.25
	Urban	170.73	0.663	169.40–172.00	0.39

Table 5.2.4 Sampling Error Measures of Total Unpaid Time by Employment Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
WLS	Employed	115.52	0.486	114.56–116.47	0.42
	Unemployed	114.61	2.631	109.45–119.76	2.30
	NILF	259.58	0.451	258.70–260.46	0.17
Bootstrap WLS	Employed	115.52	0.408	114.73–116.34	0.35
	Unemployed	114.61	2.068	110.60–118.50	1.80
	NILF	259.58	0.586	258.47–260.78	0.23

Table 5.2.5 Sampling Error Measures of Total Unpaid Time by Education

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
WLS	Illiterate	248.73	0.793	247.18–250.29	0.32
	Primary	170.34	0.527	169.31–171.38	0.31
	Secondary	191.91	0.669	190.60–193.23	0.35
	Graduation	178.62	0.993	176.67–180.56	0.56
Bootstrap WLS	Illiterate	248.73	0.883	247.04–250.52	0.35
	Primary	170.34	0.607	169.21–171.59	0.36
	Secondary	191.91	0.777	190.43–193.42	0.40
	Graduation	178.62	1.049	176.58–180.80	0.59

Table 5.2.6 Sampling Error Measures of Total Unpaid Time by Wealth index

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
WLS	Poorest	205.05	0.734	203.62–206.49	0.36
	Poor	197.42	0.761	195.93–198.91	0.39

Bootstrap WLS	Middle	193.29	0.782	191.76–194.82	0.40
	Rich	186.74	0.801	185.17–188.31	0.43
	Richest	173.11	0.807	171.53–174.69	0.47
	Poorest	205.05	0.902	203.29–206.80	0.44
	Poor	197.42	0.883	195.61–199.06	0.45
	Middle	193.29	0.865	191.61–195.00	0.45
	Rich	186.74	0.844	185.12–188.44	0.45
	Richest	173.11	0.873	171.45–174.80	0.50

Table 5.2.7 Sampling Error Measures of Total Unpaid Time by Marital Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
WLS	Never married	50.36	0.581	49.22–51.50	1.15
	Currently married	256.91	0.397	256.13–257.69	0.15
	Widowed	200.05	1.171	197.76–202.35	0.59
	Divorced/Separated	185.88	4.586	176.89–194.87	2.47
Bootstrap WLS	Never married	50.36	0.345	49.70–51.05	0.69
	Currently married	256.91	0.524	255.88–257.94	0.20
	Widowed	200.05	1.119	197.85–202.21	0.56
	Divorced/Separated	185.88	4.522	177.83–194.89	2.43

5.3. Results of Sampling Error for Time Poverty Rate

Table (5.3.1-5.3.5) The estimation of the time poverty rate using Weighted Least Squares (WLS) and Bootstrap WLS demonstrates a high degree of precision and reliability. The overall time poverty rate is estimated at 0.26, with very low standard errors (0.002–0.007) and relative standard errors (0.76–6.84%), accompanied by narrow 95% confidence intervals, indicating that the population-level estimate is robust and statistically sound.

Analysis by subgroups reveals systematic patterns consistent with socioeconomic expectations. Time poverty is higher among rural residents, less-educated individuals, the poorest wealth quintiles, and married persons, whereas lower rates are observed among urban residents, graduates, the wealthiest, and never-married individuals. Subgroups with smaller sample sizes, such as divorced or separated individuals, exhibit higher sampling variability, reflected in wider confidence intervals and elevated RSEs, although estimates remain interpretable.

Comparison of methods indicates that Bootstrap WLS produces slightly wider confidence intervals than WLS, reflecting the additional variability captured through resampling; nevertheless, the overall estimates are consistent across methods. The narrow confidence intervals and low relative standard errors across most categories confirm the robustness of the estimates and support confident inference regarding the distribution and determinants of time poverty across demographic and socioeconomic groups.

Table 5.3.1 Sampling Error Measures of Time Poverty Rate

<i>Method</i>	<i>Mean</i>	<i>SE</i>	<i>CI</i>	<i>RSE</i>
<i>WLS</i>				
<i>Time poor</i>	0.26	0.002	0.26- 0.27	0.91
<i>Bootstrap WLS</i>				
<i>Time poor</i>	0.26	0.002	0.26- 0.27	0.76

Table 5.3.2 Sampling Error Measures of Time Poverty Rate by Sector

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Rural	0.26	0.002	0.26–0.27	0.94
	Urban	0.26	0.004	0.25–0.27	1.57
<i>Bootstrap WLS</i>	Rural	0.26	0.003	0.26–0.27	1.03
	Urban	0.26	0.005	0.25–0.27	1.92

Table 5.3.3 Sampling Error Measures of Time Poverty Rate by Education

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
WLS	Illiterate	0.26	0.004	0.25–0.27	1.36
	Primary	0.27	0.004	0.27–0.28	1.31
	Secondary	0.27	0.005	0.26–0.28	1.88
	Graduation	0.23	0.006	0.22–0.24	2.60
Bootstrap WLS	Illiterate	0.26	0.004	0.25–0.27	1.56
	Primary	0.27	0.004	0.27–0.28	1.47
	Secondary	0.27	0.006	0.26–0.29	2.22
	Graduation	0.23	0.007	0.21–0.24	3.01

Table 5.3.4 Sampling Error Measures of Time Poverty Rate by Wealth Index

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
WLS	Poorest	0.29	0.005	0.28–0.30	1.57
	Poor	0.27	0.005	0.26–0.28	1.76
	Middle	0.26	0.005	0.25–0.27	1.86
	Rich	0.26	0.005	0.25–0.27	1.84
	Richest	0.24	0.005	0.23–0.24	2.03
Bootstrap WLS	Poorest	0.29	0.005	0.28–0.30	1.77
	Poor	0.27	0.005	0.26–0.28	1.95
	Middle	0.26	0.005	0.25–0.27	2.07
	Rich	0.26	0.006	0.25–0.27	2.14
	Richest	0.24	0.006	0.22–0.25	2.52

Table 5.3.5 Sampling Error Measures of Time Poverty Rate by Marital Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Never married	0.15	0.007	0.13–0.16	4.59
	Currently married	0.28	0.002	0.28–0.29	0.87
	Widowed	0.24	0.005	0.23–0.25	2.24
	Divorced/Separated	0.29	0.018	0.26–0.33	6.23
<i>Bootstrap WLS</i>	Never married	0.15	0.006	0.13–0.16	4.32
	Currently married	0.28	0.003	0.28–0.29	1.00
	Widowed	0.24	0.006	0.23–0.25	2.51
	Divorced/Separated	0.29	0.020	0.26–0.33	6.84

5.4. Results of Sampling Error for Childcare Rate

Table (5.4.1-5.4.7) The estimation of childcare participation using Weighted Least Squares (WLS) and Bootstrap WLS indicates that the overall childcare rate is 0.15, with very low standard errors (0.002) and relative standard errors (1.17–1.30%), accompanied by narrow 95% confidence intervals (0.15–0.16). This reflects a highly precise estimate of the population-level childcare rate.

Subgroup analyses reveal notable demographic and socioeconomic patterns. Females, rural residents, illiterate individuals, the poorest households, and widowed individuals exhibit higher childcare rates, whereas males, urban residents, graduates, the richest households, and never-married individuals report lower rates. Subgroups with smaller sample sizes, such as the unemployed or divorced/separated individuals, show higher standard errors, wider confidence intervals, and elevated RSEs (up to 25.81%), indicating greater sampling variability, although the estimates remain interpretable.

Comparison between methods indicates that Bootstrap WLS produces slightly wider confidence intervals and marginally higher RSEs, reflecting the additional variability captured through resampling. Overall, the narrow confidence intervals and generally low relative standard errors across most categories confirm the robustness of the estimates, supporting

reliable inference regarding the distribution of childcare responsibilities across demographic, socioeconomic, and marital groups.

Table (5.4.1-5.4.7)

Table 5.4.1 Sampling Error Measures of Childcare Rate

<i>Method</i>	<i>Mean</i>	<i>SE</i>	<i>CI</i>	<i>RSE</i>
<i>WLS</i>				
<i>Childcare</i>	0.15	0.002	0.15- 0.16	1.3
<i>Bootstrap WLS</i>				
<i>Childcare</i>	0.15	0.002	0.15- 0.16	1.17

Table 5.4.2 Sampling Error Measures of Childcare Rate by Gender

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Male	0.12	0.002	0.11–0.12	1.87
	Female	0.19	0.002	0.19–0.19	1.17
<i>Bootstrap WLS</i>	Male	0.12	0.002	0.11–0.12	1.88
	Female	0.19	0.003	0.18–0.19	1.42

Table 5.4.3 Sampling Error Measures of Childcare Rate by Sector

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Rural	0.16	0.002	0.15–0.16	1.22
	Urban	0.15	0.003	0.15–0.16	1.88
<i>Bootstrap WLS</i>	Rural	0.16	0.002	0.15–0.16	1.37
	Urban	0.15	0.003	0.15–0.16	2.06

Table 5.4.4 Sampling Error Measures of Childcare Rate by Employment Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Employed	0.13	0.003	0.12–0.13	2.10
	Unemployed	0.21	0.045	0.12–0.30	21.32
	NILF	0.17	0.002	0.17–0.17	1.16
<i>Bootstrap WLS</i>	Employed	0.13	0.003	0.12–0.13	2.06
	Unemployed	0.21	0.052	0.11–0.31	24.91
	NILF	0.17	0.002	0.16–0.17	1.36

Table 5.4.5 Sampling Error Measures of Childcare Rate by Education

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Illiterate	0.17	0.002	0.17–0.18	1.32
	Primary	0.15	0.003	0.14–0.15	1.90
	Secondary	0.13	0.004	0.12–0.14	3.26
	Graduation	0.10	0.006	0.09–0.12	5.85
<i>Bootstrap WLS</i>	Illiterate	0.17	0.003	0.17–0.18	1.57
	Primary	0.15	0.003	0.14–0.15	2.07
	Secondary	0.13	0.004	0.12–0.14	3.39
	Graduation	0.10	0.006	0.09–0.12	5.35

Table 5.4.6 Sampling Error Measures of Childcare Rate by Wealth Index

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
<i>WLS</i>	Poorest	0.24	0.003	0.23–0.24	1.39
	Poor	0.19	0.003	0.18–0.20	1.78

Bootstrap WLS	Middle	0.15	0.003	0.14–0.16	2.34
	Rich	0.11	0.004	0.10–0.11	3.35
	Richest	0.07	0.004	0.06–0.07	5.61
	Poorest	0.24	0.004	0.23–0.25	1.70
	Poor	0.19	0.004	0.18–0.20	2.25
	Middle	0.15	0.004	0.14–0.16	2.60
	Rich	0.11	0.003	0.10–0.11	3.26
	Richest	0.07	0.003	0.06–0.07	4.96

Table 5.4.7 Sampling Error Measures of Childcare Rate by Marital Status

<i>Method</i>	<i>Category</i>	<i>Mean</i>	<i>SE</i>	<i>95% CI</i>	<i>RSE (%)</i>
WLS	Never married	0.07	0.018	0.03–0.10	25.81
	Currently married	0.14	0.002	0.14–0.15	1.38
	Widowed	0.18	0.003	0.17–0.19	1.50
	Divorced/Separated	0.13	0.023	0.09–0.17	17.33
Bootstrap WLS	Never married	0.07	0.014	0.04–0.10	20.14
	Currently married	0.14	0.002	0.14–0.15	1.49
	Widowed	0.18	0.003	0.17–0.19	1.81
	Divorced/Separated	0.13	0.023	0.08–0.17	17.88

5.5. Discussion

The analysis of sampling errors demonstrates that the estimates of total unpaid time, time poverty rate, and childcare rate are highly precise and reliable, with low standard errors, narrow confidence intervals, and generally low relative standard errors, indicating minimal sampling error for the overall population. Subgroup analysis highlights that smaller or less-represented groups, such as the unemployed, divorced/separated, and never-married individuals, exhibit higher sampling variability, reflected in wider confidence intervals and elevated RSEs. This emphasizes the importance of considering sampling error when interpreting estimates for these subpopulations. Comparisons between WLS and Bootstrap WLS show that Bootstrap WLS generally produces slightly wider confidence intervals, capturing additional resampling variability, yet both methods yield consistent and robust estimates. Overall, the low sampling variability across most categories confirms the reliability of the survey data and supports confident interpretation of patterns in unpaid work, time poverty, and childcare responsibilities.

CHAPTER 6

CONCLUSION

6.1. Comparative Discussion of Variance Estimation Methods

Within design-based methods, each approach demonstrates distinct strengths depending on sample size, complexity, and type of statistic. Taylor Linearization performs efficiently for large samples and standard survey indicators, providing precise variance estimates for totals and averages, but requires linearization of the variable, which may not always be feasible. Jackknife and Bootstrap are replication-based methods that do not rely on linearization, making them more flexible for small samples, complex statistics, and rate-type variables. Both capture the impact of clustering and unequal weights, and their repeated resampling enables accurate estimation of variability even in heterogeneous or sparsely represented subgroups. Design effects (DEFF) are particularly informative in design-based methods, quantifying how stratification, clustering, or unequal weights inflate variance relative to simple random sampling. Higher DEFF values, observed in subgroups such as males or employed individuals, highlight the increased variability induced by the survey design, guiding careful interpretation of subgroup estimates.

Comparing design-based and model-based methods, the main distinction lies in the source of variability considered. Design-based methods rely solely on the sampling design, producing variance estimates that reflect the survey structure and weighting, while model-based methods, such as Weighted Least Squares (WLS) and Bootstrap WLS, combine survey weights with statistical modelling to produce efficient and precise estimates of means, totals, and continuous outcomes. Model-based methods are particularly advantageous when resampling is incorporated, as in Bootstrap WLS, capturing additional variability not reflected by standard WLS. However, model-based methods assume correct model specification including linearity, independence, and homoscedasticity and can underestimate variance if assumptions are violated. In contrast, design-based methods remain robust regardless of model assumptions, with DEFF providing explicit information about the influence of clustering and weighting on precision.

Overall, design-based methods offer robustness and transparency, ensuring that variance estimates account for the actual survey design and highlighting areas of high sampling

variability through DEFF. Model-based methods provide efficiency and stability, especially for weighted analyses and small samples, but rely on model assumptions. Using both approaches in tandem allows for a comprehensive assessment of sampling variability, balancing reliability, precision, and interpretability in survey analysis.

6.2. Strengths of the Study

- The study utilizes a nationally representative time use survey, enhancing the generalizability of findings to other surveys with comparable designs.
- Survey estimates demonstrate robustness and stability, with consistency observed across both design-based and model-based variance estimation methods.
- A comprehensive approach is employed, incorporating Taylor Linearization, Jackknife, Bootstrap, WLS, and Bootstrap WLS, providing a thorough assessment of variance, standard errors, and sampling variability.
- The inclusion of design effect (DEFF) analysis allows for the evaluation of how clustering and stratification influence variance estimates, highlighting the impact of complex survey design on the precision of estimates.
- Methods are applied directly to empirical survey data, illustrating their practical applicability and effectiveness in capturing variability.
- The resulting estimates exhibit high precision, evidenced by small standard errors, low relative standard errors, narrow confidence intervals.

6.3. Limitations of the Study

- Balanced Repeated Replication (BRR) could not be implemented, as it requires precisely two primary sampling units per stratum, a condition not met in large-scale surveys in India.
- Model-based methods rely on assumptions including linearity, independence, homoscedasticity, and normality of residuals; violations of these assumptions may lead to biased variance estimates.
- Replication-based methods, such as Jackknife and Bootstrap, while robust, are computationally intensive, particularly when a large number of replicates is required.

- The selection of an appropriate variance estimation method necessitates careful consideration of precision, survey design, computational feasibility, and statistical complexity, which can present practical challenges in certain contexts.

6.4. Conclusion

Accurate estimation of variance and sampling error is fundamental to high-quality survey research because it directly influences the precision and credibility of findings (Binder & Roberts, in *Analysis of Survey Data*; Rao & Wu, *JASA*). Design-based methods explicitly account for complex survey structures such as stratification, clustering, and unequal weights, and the interpretation of design effects helps quantify how these features inflate variance compared to simple random sampling, thus guiding correct standard error estimation and confidence interval construction. Model-based approaches offer efficient alternatives for weighted or continuous outcomes when model assumptions are met, but they still require careful evaluation to avoid underestimating variability. The design effect (DEFF) provides valuable insight into how the survey design influences precision, highlighting where sampling variability is amplified due to clustering or weighting. Therefore, survey researchers must select appropriate variance estimation methods and interpret standard errors in light of the survey design, as correct understanding of sampling variability ensures that survey results are robust, credible, and interpretable within the context of the underlying sampling process.

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APPENDIX 1 (International Classification of Activities for Time Use Statistics 2016 (ICATUS 2016))

Activity and code	Activity and code
1 Employment and related activities	223 Wood products
11 Employment in corporations, government and non-profit institutions	224 Construction materials
110 Employment in corporations, government and non-profit institutions	225 Herbal products
12 Employment in household enterprises to produce goods	226 Metal products
121 Growing of crops for the market	227 Other materials
122 Raising animals for the market	229 Related activities
123 Forestry and logging for the market	23 Construction for own use
124 Fishing for the market	230 Construction
125 Aquaculture for the market	24 Water and fuel collection
126 Mining and quarrying for the market	241 Firewood collection
127 Making and processing goods for the market	242 Fetching water
128 Construction activities for the market	25 Travel related to own production
129 Other activities related to employment (goods)	250 Transport activities
13 Employment in household enterprises to provide services	3 Unpaid domestic services
131 Vending and trading	31 Food preparation
132 Repair and maintenance services	311 Preparing meals
133 Professional services	312 Serving meals
134 Transport services	313 Cleaning after meals
135 Personal care services	314 Food storage
136 Paid domestic services	319 Other food activities
139 Other services	32 Cleaning and maintenance
14 Ancillary activities and breaks	321 Indoor cleaning
141 Ancillary activities	322 Outdoor cleaning
142 Breaks	323 Garbage disposal
15 Training and studies	324 Gardening
150 Training related to employment	329 Other cleaning
16 Seeking employment	33 Household repair
160 Job search	331 House repair
17 Setting up business	332 Equipment repair
170 Business setup	333 Vehicle repair
18 Travel for employment	339 Other repair
181 Work travel	34 Textiles care
182 Commuting	341 Washing
2 Production of goods for own final use	342 Drying
21 Agriculture for own use	343 Ironing
211 Growing crops	344 Repair clothes
212 Animal farming	349 Other textile care
213 Hunting and trapping	35 Household management
214 Forestry	351 Paying bills
215 Gathering wild products	352 Planning activities
216 Fishing	359 Other management
217 Aquaculture	36 Pet care
218 Mining	361 Daily pet care
22 Making and processing goods	362 Veterinary care
221 Food processing	369 Other pet care
	37 Shopping
	371 Goods purchase
	372 Services purchase
	38 Travel domestic services
	380 Transport activities

222 Textile production	620 Homework/study
4 Unpaid caregiving	63 Non-formal education
41 Childcare	630 Courses
411 Physical care	7 Social and community activities
412 Medical care	71 Socializing
413 Teaching	711 Talking
414 Talking/reading	712 Gathering
415 Playing	713 Communication
416 Passive care	719 Other
417 Meetings	72 Community events
419 Other childcare	721 Cultural events
42 Adult care	722 Social rites
421 Daily assistance	723 Functions
422 Medical care	729 Other
423 Administrative help	73 Civic duties
424 Emotional support	730 Civic activities
425 Passive care	74 Religion
426 Meetings	741 Prayer
429 Other adult care	742 Collective practice
43 Help to non-dependent adults	749 Other
431 Physical care	8 Leisure and sports
432 Emotional support	81 Events
439 Other help	811 Cultural events
44 Travel caregiving	812 Parks
441 Travel	813 Sports events
442 Accompanying children	819 Other
443 Accompanying adults	82 Hobbies
444 Accompanying others	821 Arts
5 Unpaid volunteer work	822 Hobbies
51 Volunteer for households	823 Games
511 Maintenance	829 Other
512 Shopping	83 Sports
513 Childcare	831 Sports
514 Adult care	832 Exercise
515 Enterprise help	84 Media
519 Other volunteering	841 Reading
52 Community volunteering	842 TV
521 Construction/cleaning	843 Radio
522 Meal services	849 Other
523 Cultural activities	85 Relaxation
524 Administrative work	850 Resting
529 Other	9 Self-care
53 Trainee work	91 Sleep
530 Trainee activities	911 Night sleep
54 Travel volunteering	912 Naps
540 Travel	913 Sleeplessness
6 Learning	919 Other
61 Formal education	92 Eating
611 School attendance	921 Meals
612 Extra activities	922 Drinking
613 Breaks	93 Hygiene
614 Self-study	931 Personal care
619 Other 62 Homework	932 Medical care 939 Other